



北京大学
PEKING UNIVERSITY

Energy- and Area-efficient Fe-FinFET-based Time-Domain Mixed-Signal Computing In Memory for Edge Machine Learning

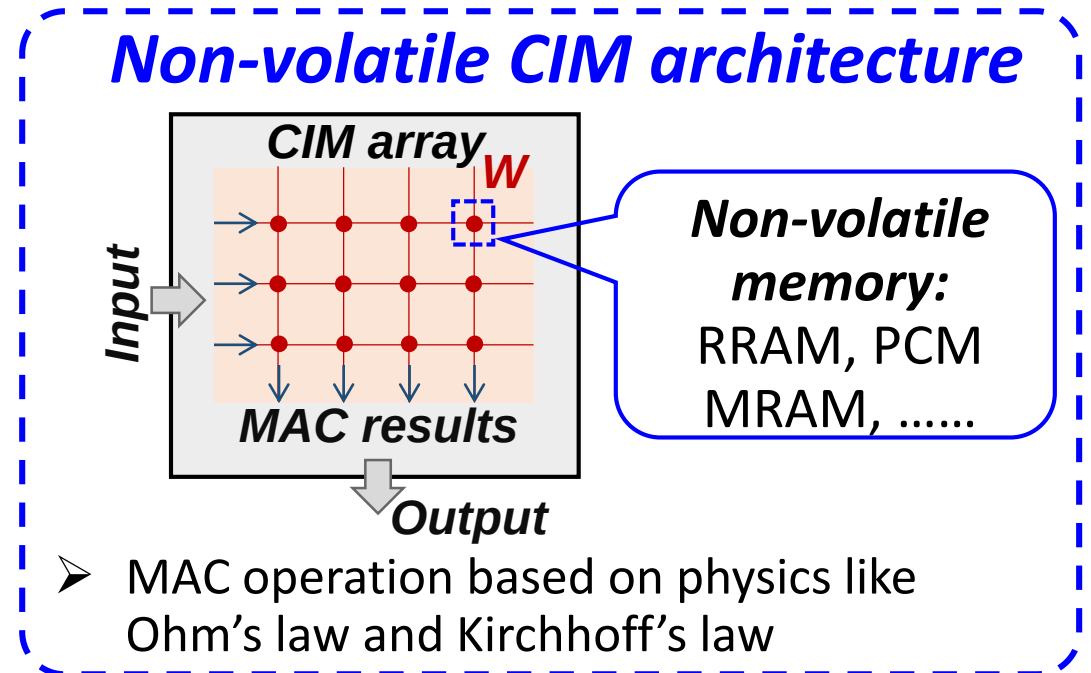
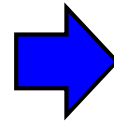
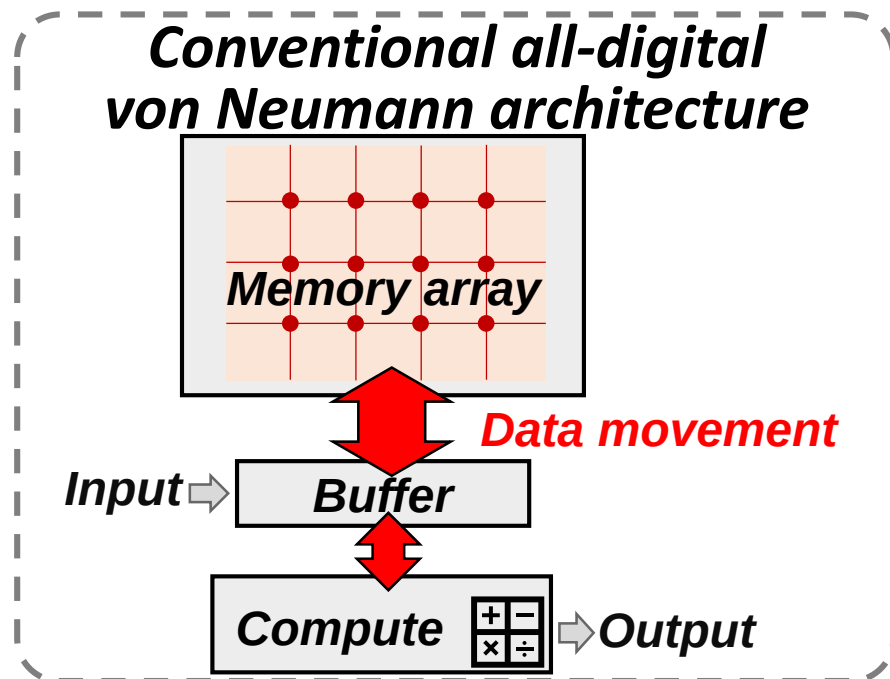
Jin Luo, Weikai Xu, Yide Du, Boyi Fu, Jiahao Song, Zhiyuan Fu,
Mengxuan Yang, Yiqing Li, Le Ye, Qianqian Huang* and Ru Huang*

*School of Integrated Circuits,
Peking University, Beijing 100871, China*

Motivation (1/3)

Non-volatile computing-in-memory for edge AI

- The conventional Von Neumann architecture: **memory wall** bottleneck
- Non-volatile memory based computing-in-memory (nvCIM) architecture
 - **Highly-parallel multiply-accumulate (MAC)** inside memory array
 - **Non-volatile device** based weight storage (RRAM, PCM, MRAM, etc.)



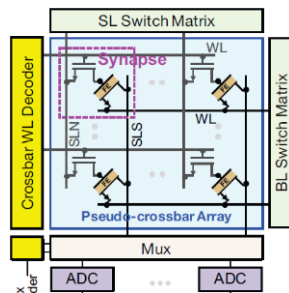
Motivation (2/3)

Ferroelectric FET based nvCIM

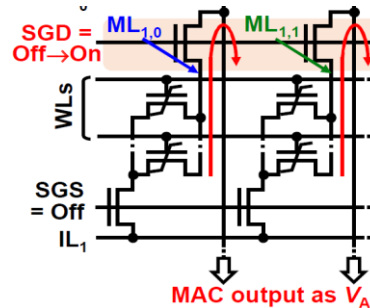
- Ferroelectric FET (FeFET) for nvCIM
 - Non-volatile ferroelectric (FE) polarization for multilevel weight storage
 - Channel conductance modulation for MAC operation
- MAC signal expression in **voltage domain**
 - Accommodate signal dynamic range with **high operating voltage**
 - **Additional data-conversion analog blocks**

FeFET based nvCIM:

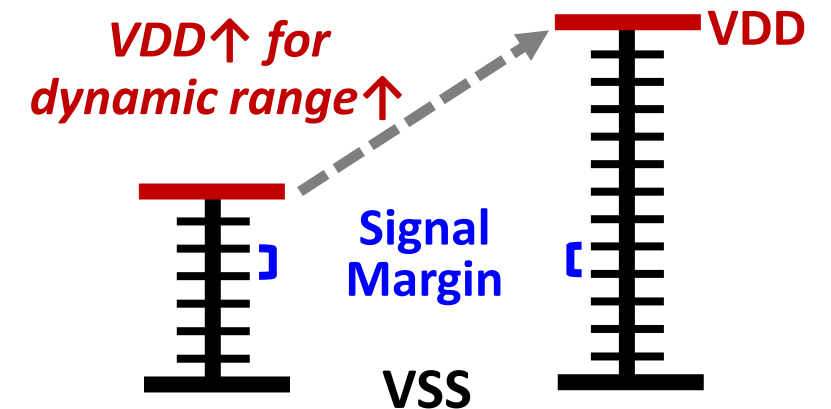
MAC signal is expressed in voltage domain



Current integration mode
M. Jerry et al., IEDM, 2017



Charge sharing mode
C. Matsui et al., VLSI, 2021.

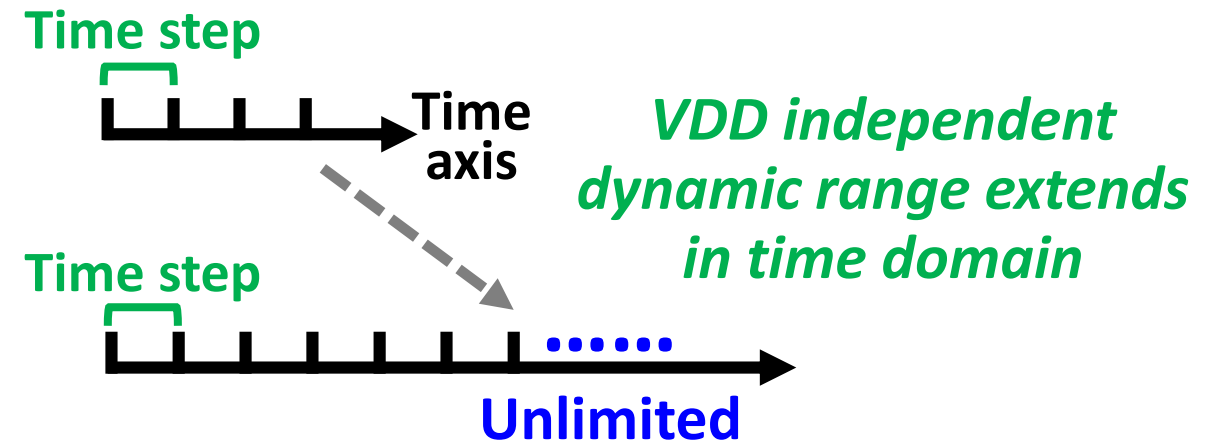
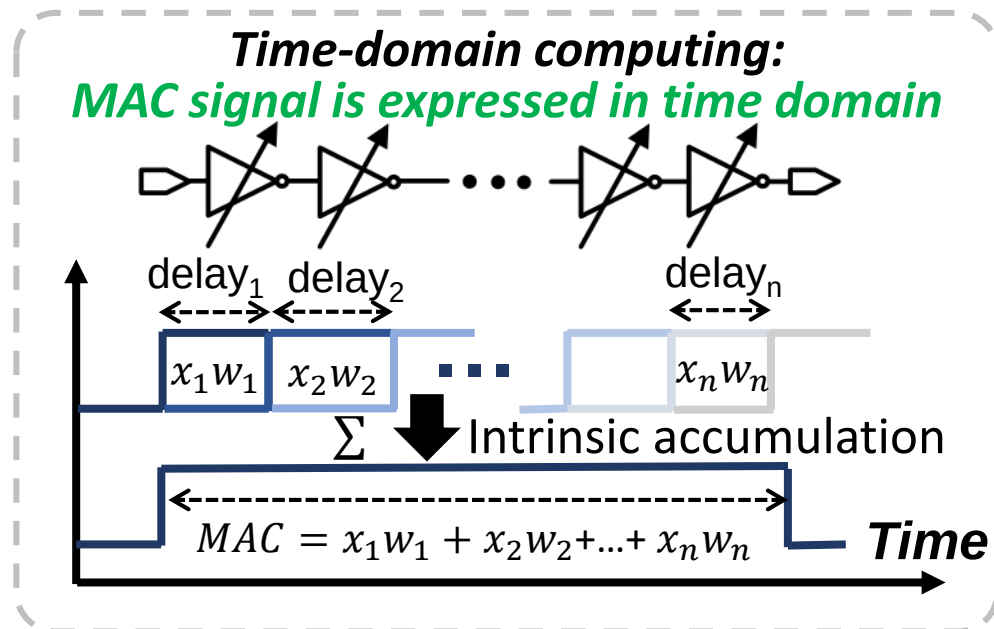


$$V_{\text{output swing}} = V_{\text{signal margin}} \times \#_{\text{MAC}}$$

Motivation (3/3)

Time-domain computing

- Time-domain (TD) MAC
 - Utilize cumulative propagation delay in circuit (e. g. cascaded inverter chain)
 - Signal dynamic range in time-domain, **independent with VDD**
 - **Digital design compatibility** w/o extra analog design
- } **Enable energy-efficiency improvement**

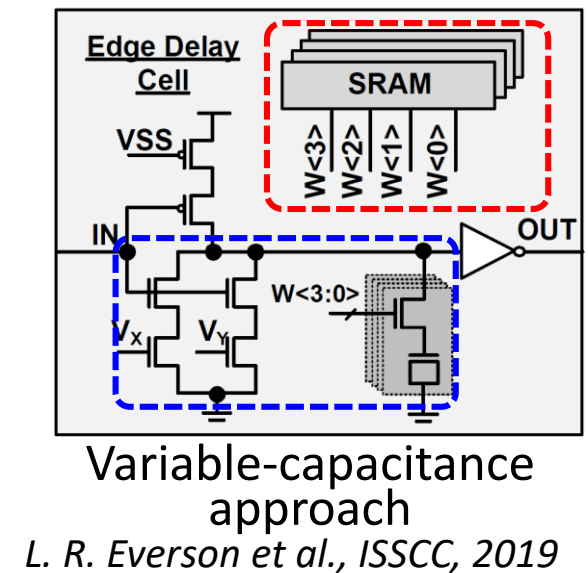
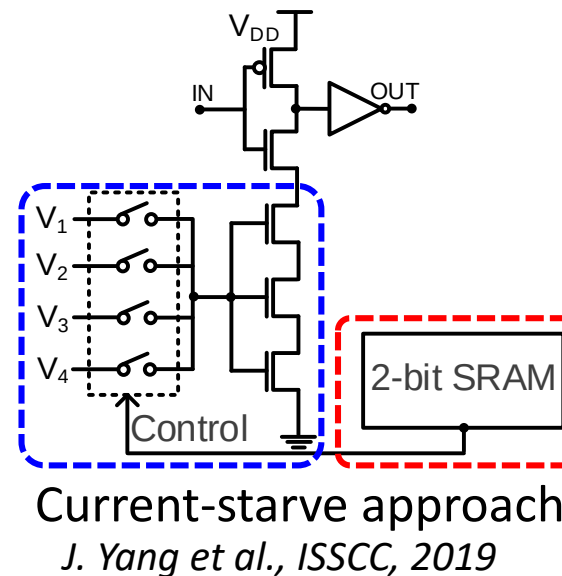
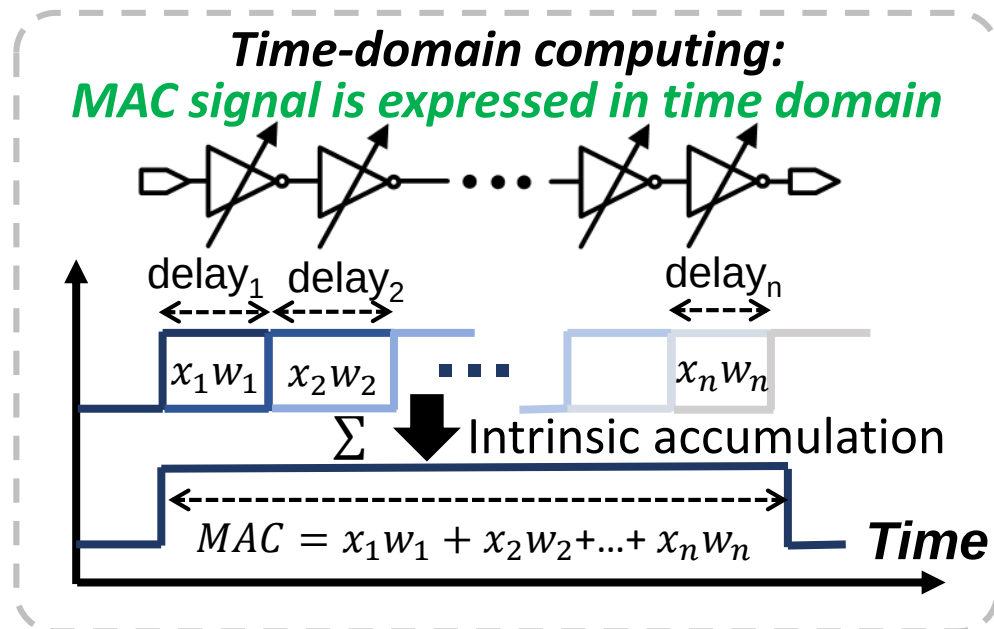


Motivation (3/3)

Time-domain computing

- Time-domain (TD) MAC
- Hardware implementation of TD MAC
 - **Separated** weight storage and multiplication operation
 - Weight storage based on SRAM & modulation circuit for multiplication

Suffer from
additional energy
and area cost



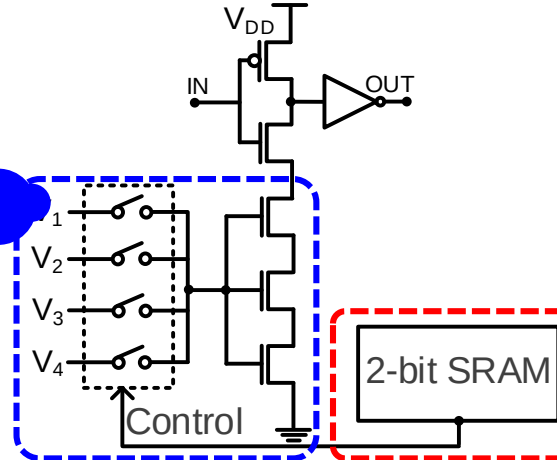
Motivation (3/3)

Time-domain computing

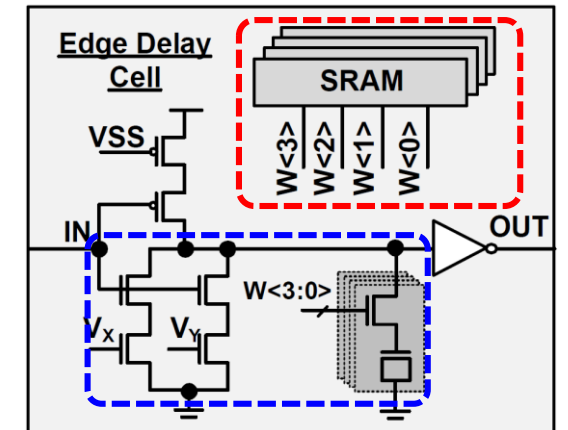
- Time-domain (TD) MAC
- Hardware implementation of TD MAC
 - **Separated** weight storage and multiplication operation
 - Weight storage based on SRAM & modulation circuit for multiplication

Suffer from
additional energy
and area cost

*Can weight storage
and multiplication
be simultaneously
implemented in
a compact device?*



Current-starve approach
J. Yang et al., ISSCC, 2019



Variable-capacitance
approach
L. R. Everson et al., ISSCC, 2019

In this work of FeFET-based TD nvCIM

- Implement weight and multiplication simultaneously in one FeFET for time-domain MAC based on advanced technology node and scaled VDD with high energy- & area-efficiency

FeFET-based delay unit (DU) with **2T+1FeFET**

FE spontaneous polarization

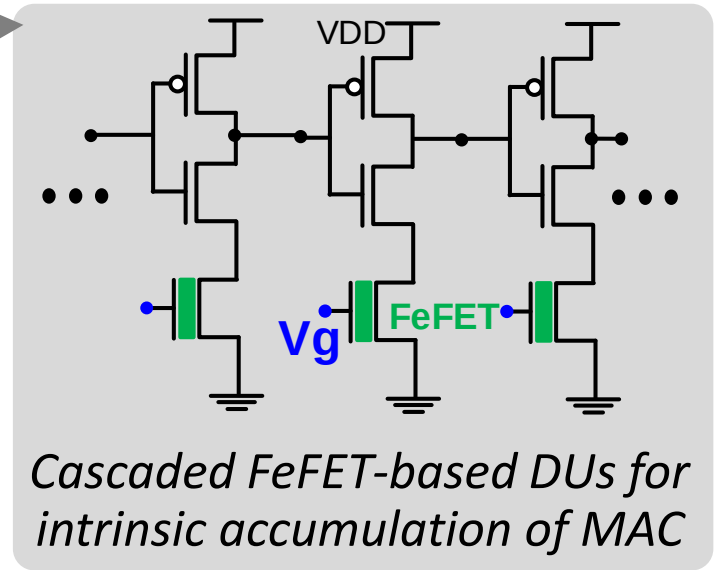
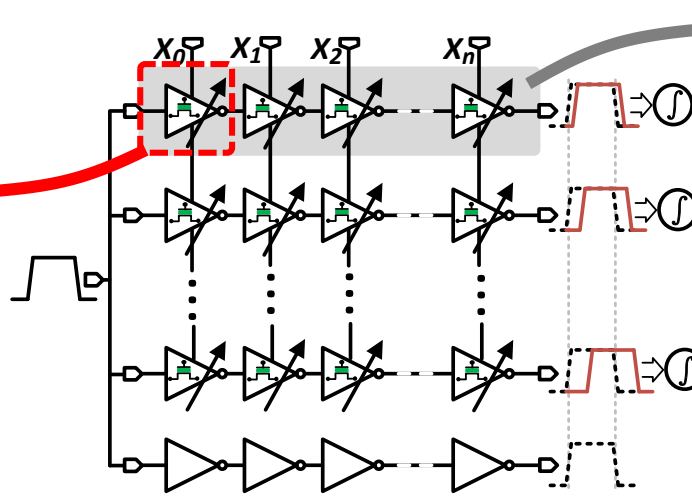


Non-volatile weight storage

Controllable switch of three-terminal FeFET



Local multiplication of MAC



In this work of FeFET-based TD nvCIM

- Implement weight and multiplication simultaneously in one FeFET for time-domain MAC based on **advanced technology node** and **scaled VDD** with high energy- & area-efficiency

FeFET-based delay unit (DU) with **2T+1FeFET**

FE spontaneous polarization

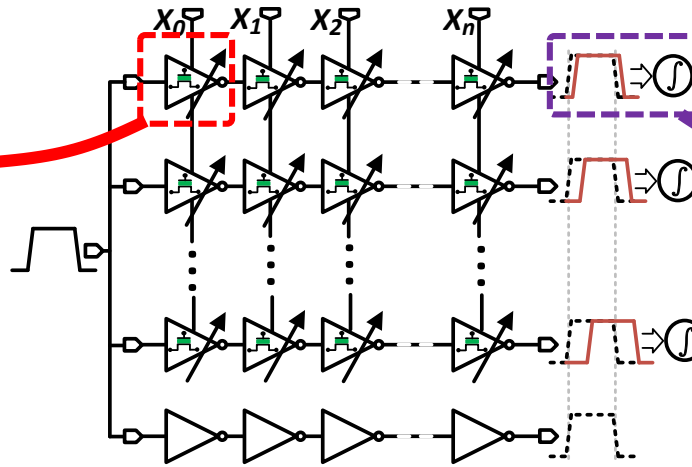


Non-volatile weight storage

Controllable switch of three-terminal FeFET



Local multiplication of MAC



TD activation with only **1FeFET**

Time-dependent FE polarization switching



TD activation function with memory trace

In this work of FeFET-based TD nvCIM

- Implement weight and multiplication simultaneously in one FeFET for time-domain MAC based on advanced technology node and scaled VDD with high energy- & area-efficiency

FeFET-based delay unit (DU) with **2T+1FeFET**

FE spontaneous polarization

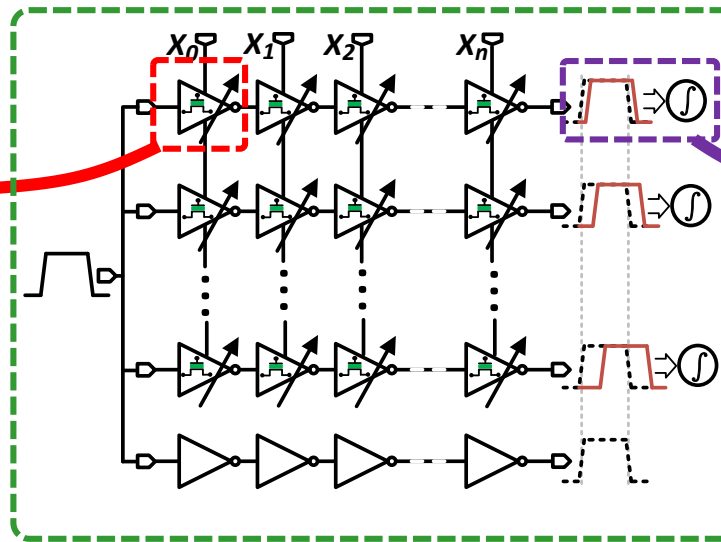


Non-volatile weight storage

Controllable switch of three-terminal FeFET



Local multiplication of MAC



TD activation with only **1FeFET**

Time-dependent FE polarization switching



TD activation function with memory trace

Edge AI based on proposed FeFET-based TD nvCIM

- ✓ Pattern recognition & reinforcement learning
- ✓ **High energy- and area-efficiency**

Outline

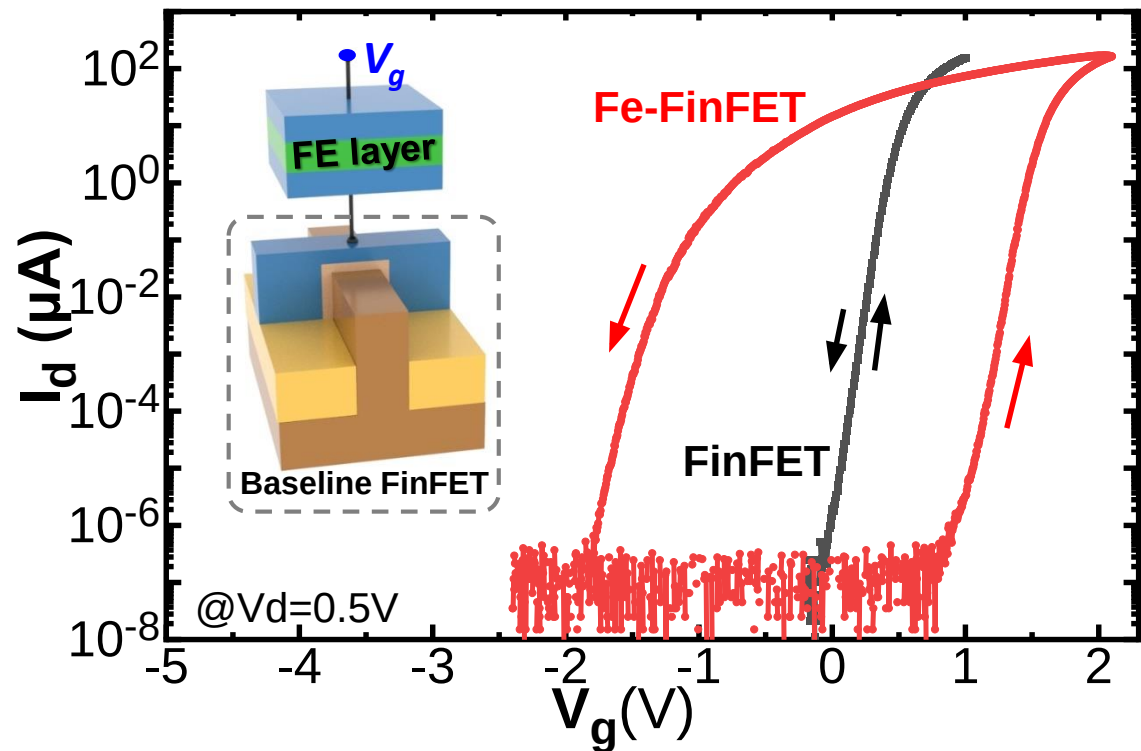
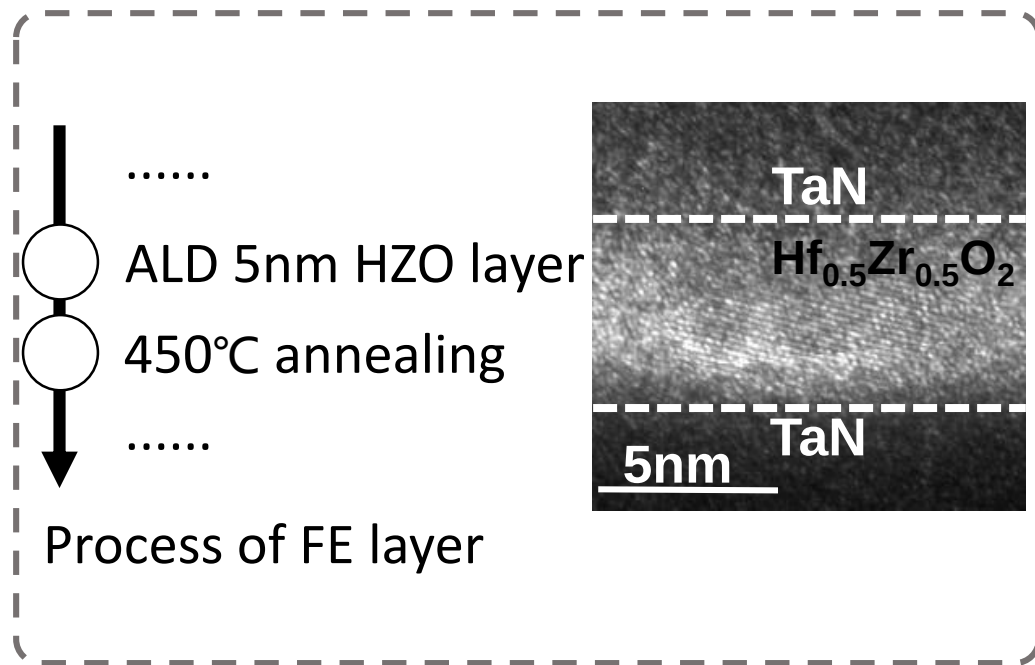
- Design of FeFET-based time-domain (TD) MAC
- FeFET-based delay unit (DU) for TD MAC
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge application
- Summary

Outline

- Design of FeFET-based time-domain (TD) MAC
- FeFET-based delay unit (DU) for TD MAC
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge application
- Summary

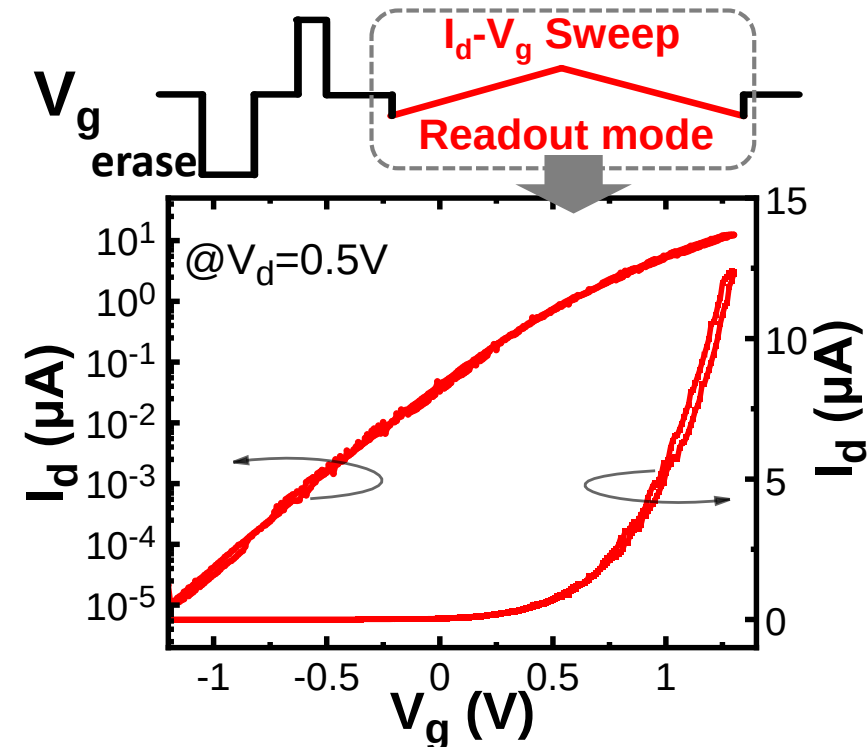
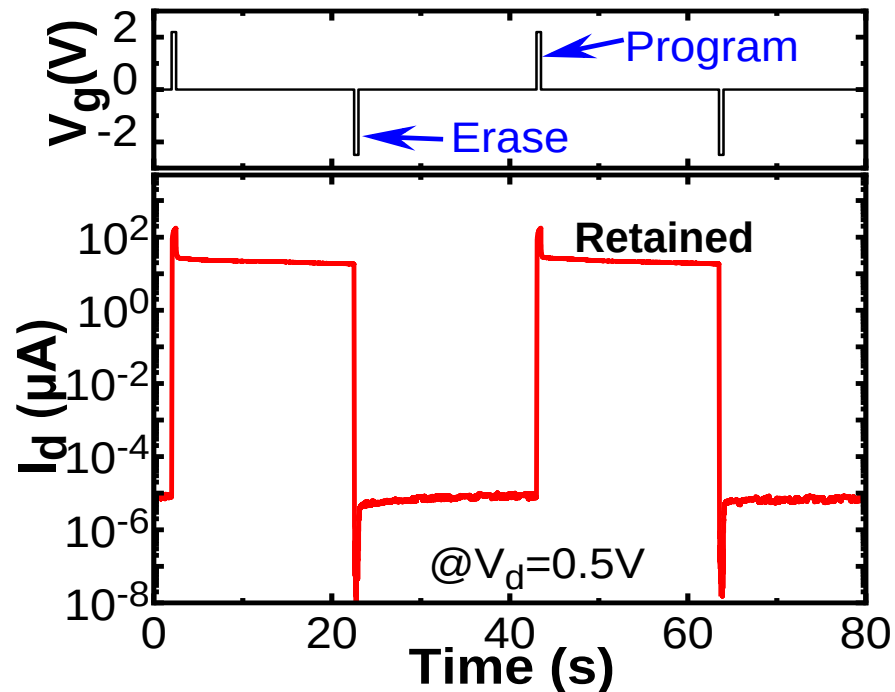
Features of basic FeFET (1/2)

- The experimental FeFET is fabricated based on FinFET structure
 - I_d - V_g characteristics show typical counterclockwise hysteresis



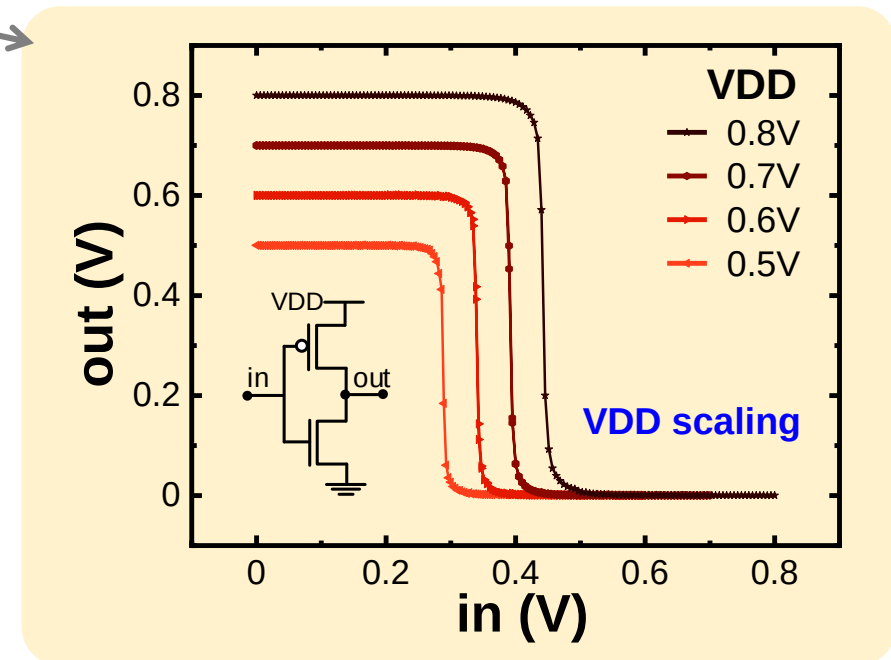
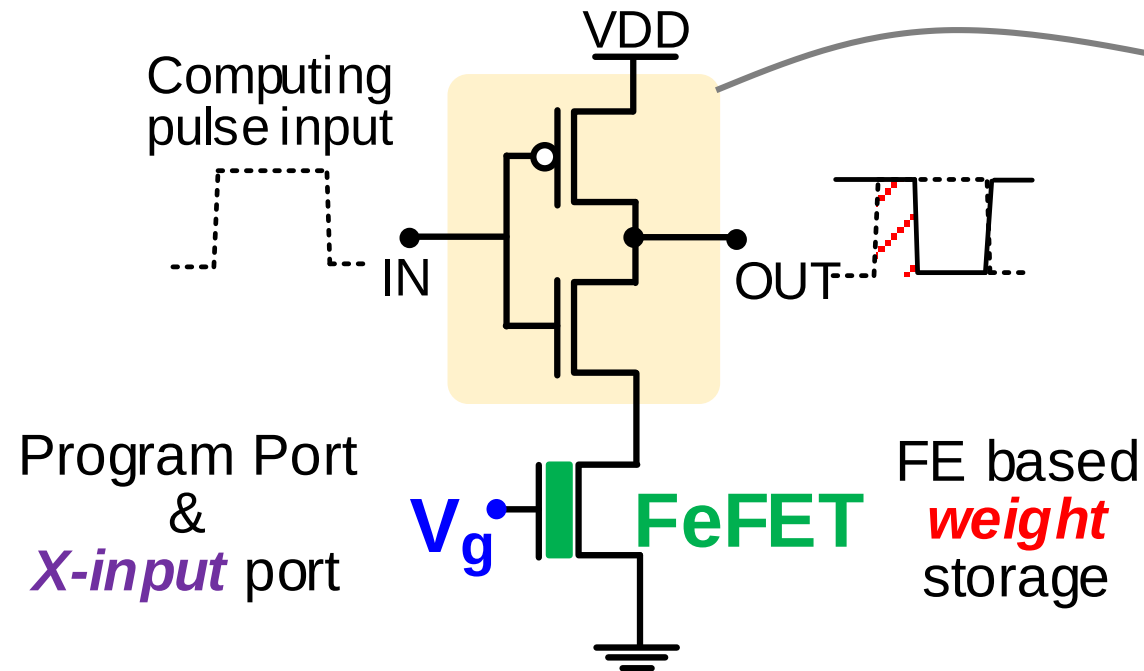
Features of basic FeFET (2/2)

- Program/erase mode with large $|V_g|$
 - Non-volatile ferroelectric (FE) polarization behavior
- Non-destructive readout mode with small $|V_g|$
 - I_d - V_g curve shows negligible hysteresis and large I_{ON}/I_{OFF} ratio



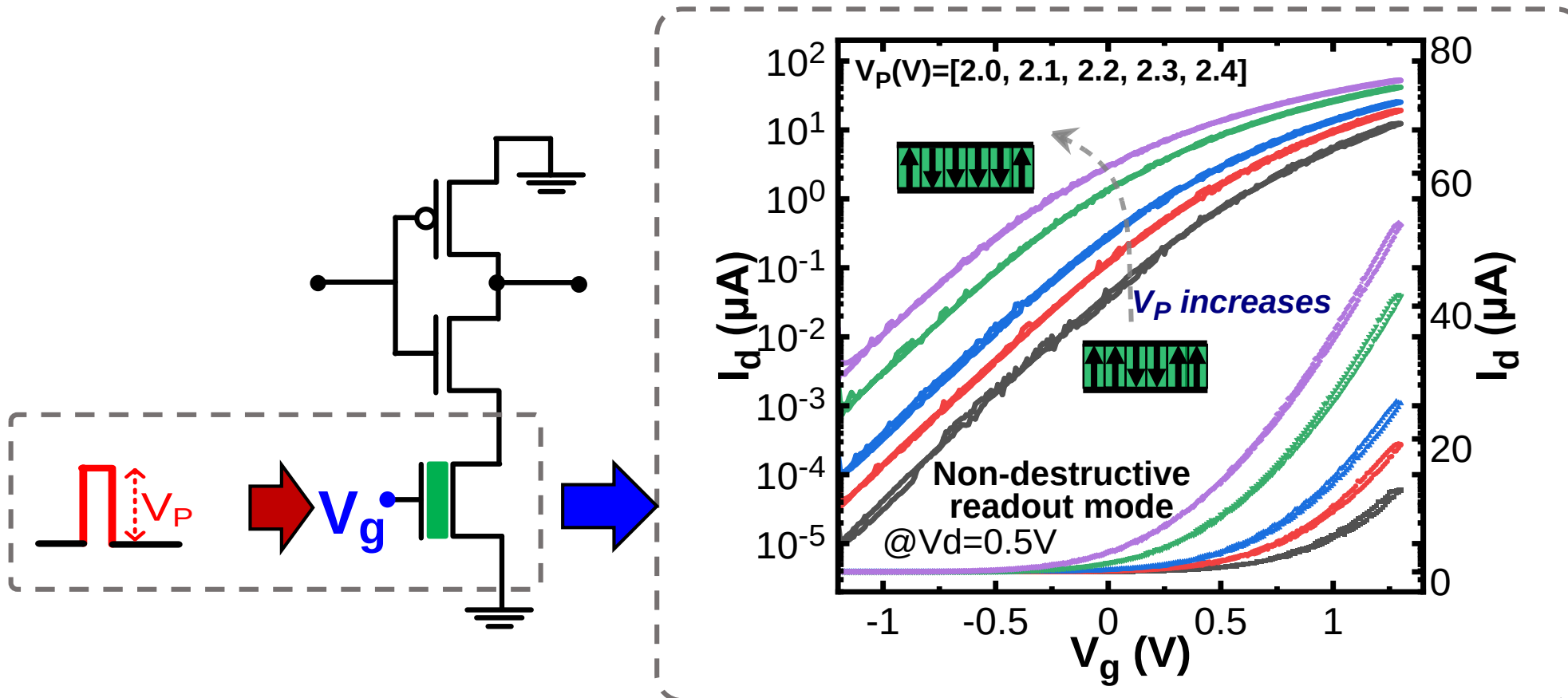
FeFET-based delay unit design (1/5)

- Topology design of FeFET-based delay unit (FE-DU) with **2T+1FeFET**
 - Fe-FinFET is used to simultaneously implement weight storage and multiplication
 - Complementary FinFET based inverter



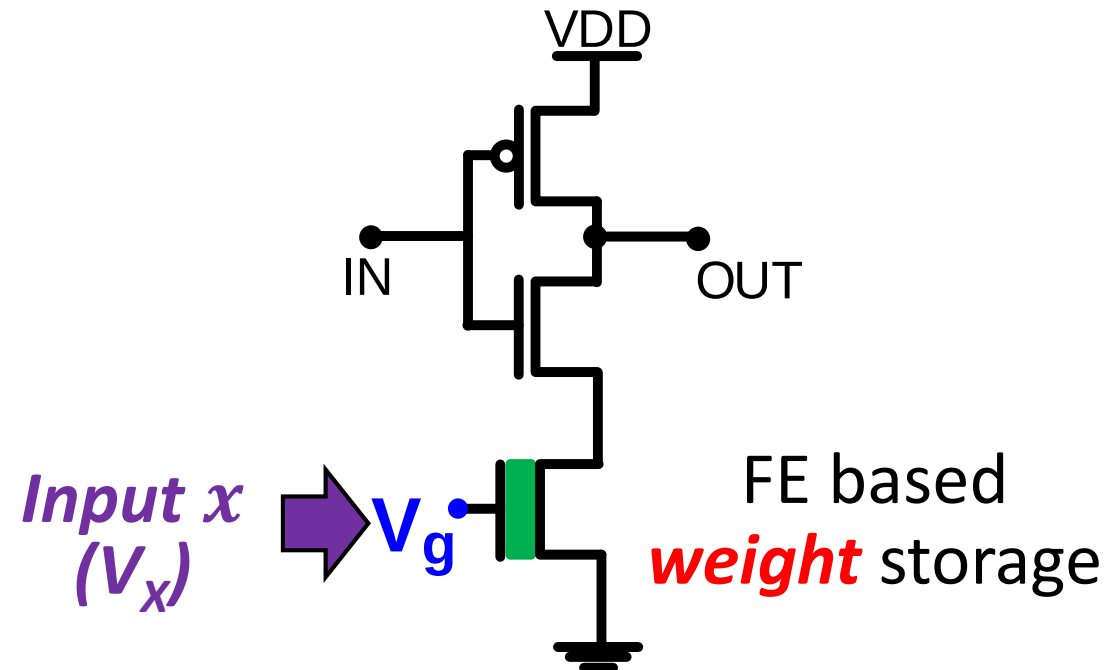
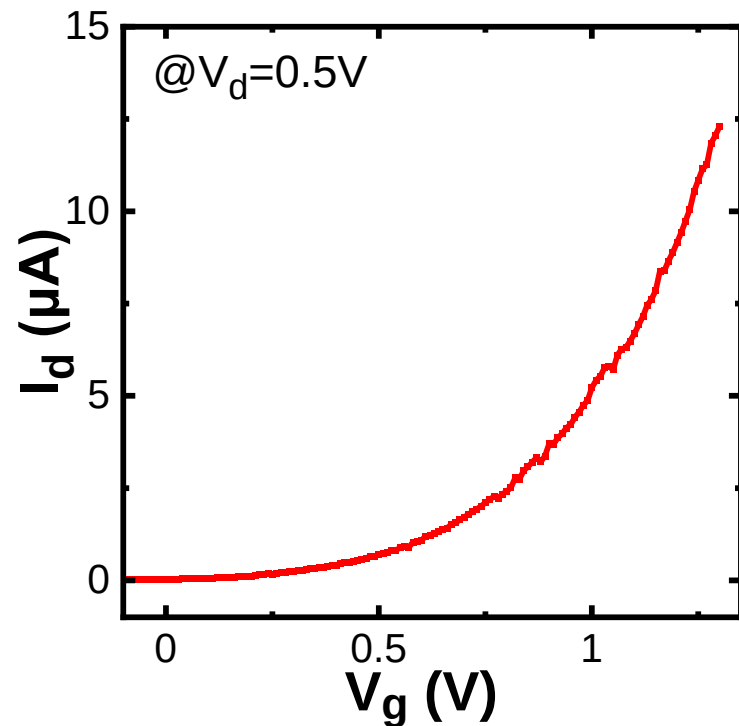
FeFET-based delay unit design (2/5)

- The programmable weight (w) is stored in FE polarization
 - Program voltage (V_P) with high amplitude to adjust FE polarization states



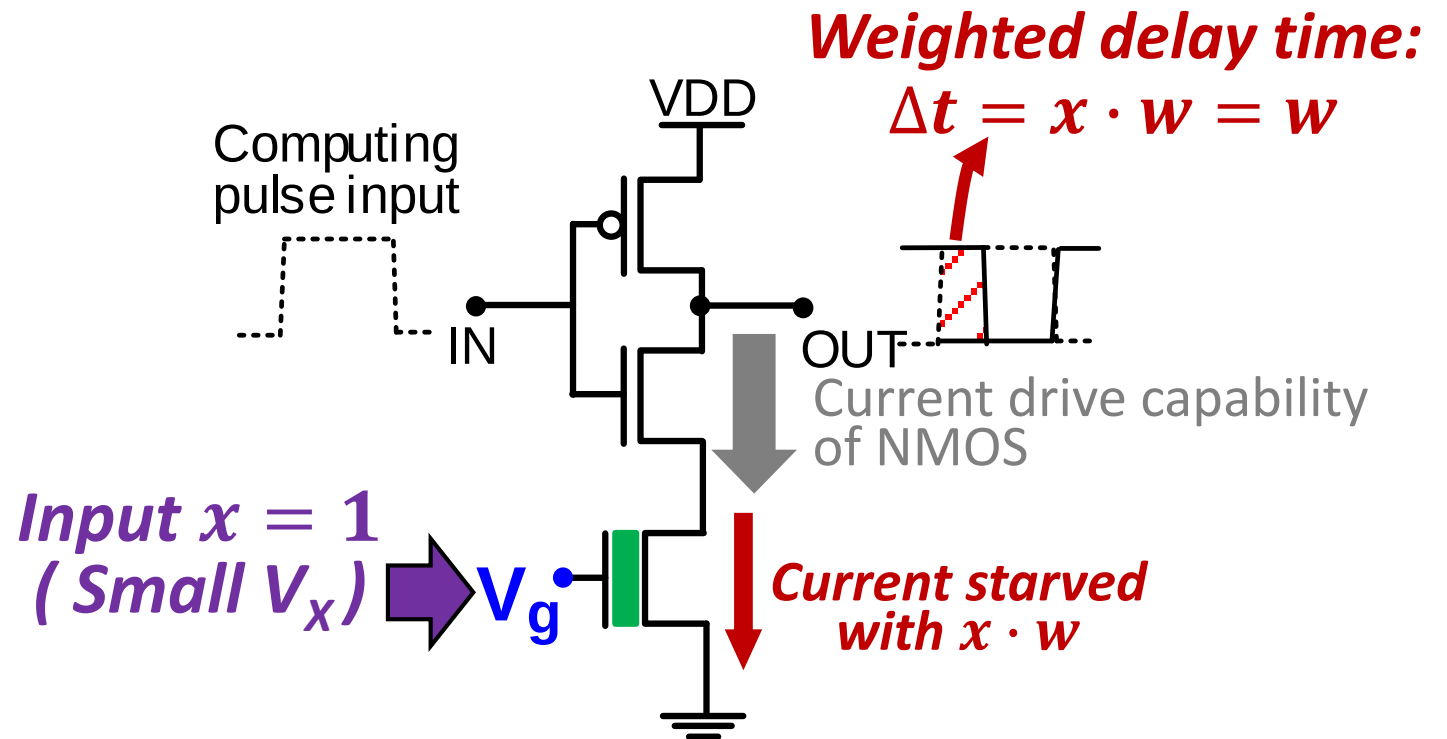
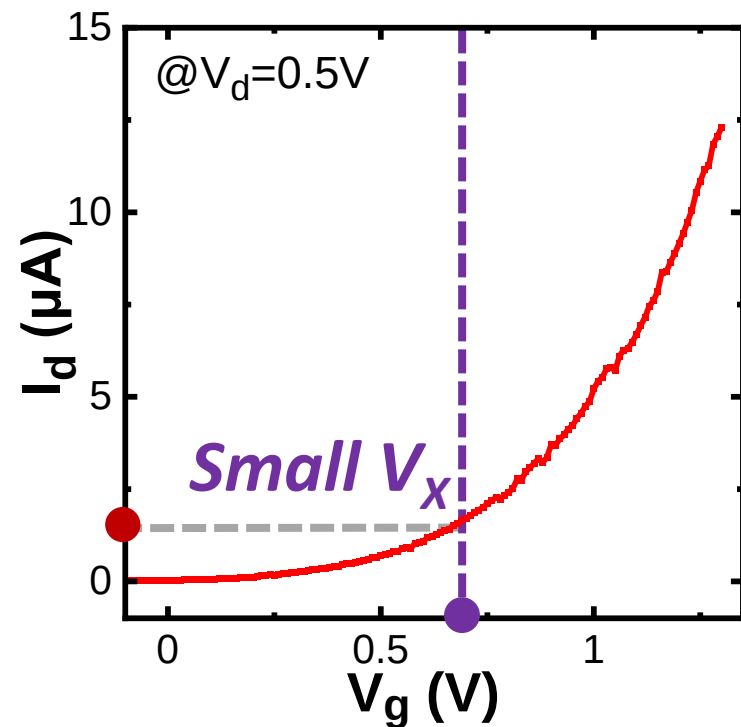
FeFET-based delay unit design (3/5)

- Drive current of FeFET is determined by weight and V_g in readout mode
 - ***In-situ* multiplication** of weight w & input data x in the FeFET



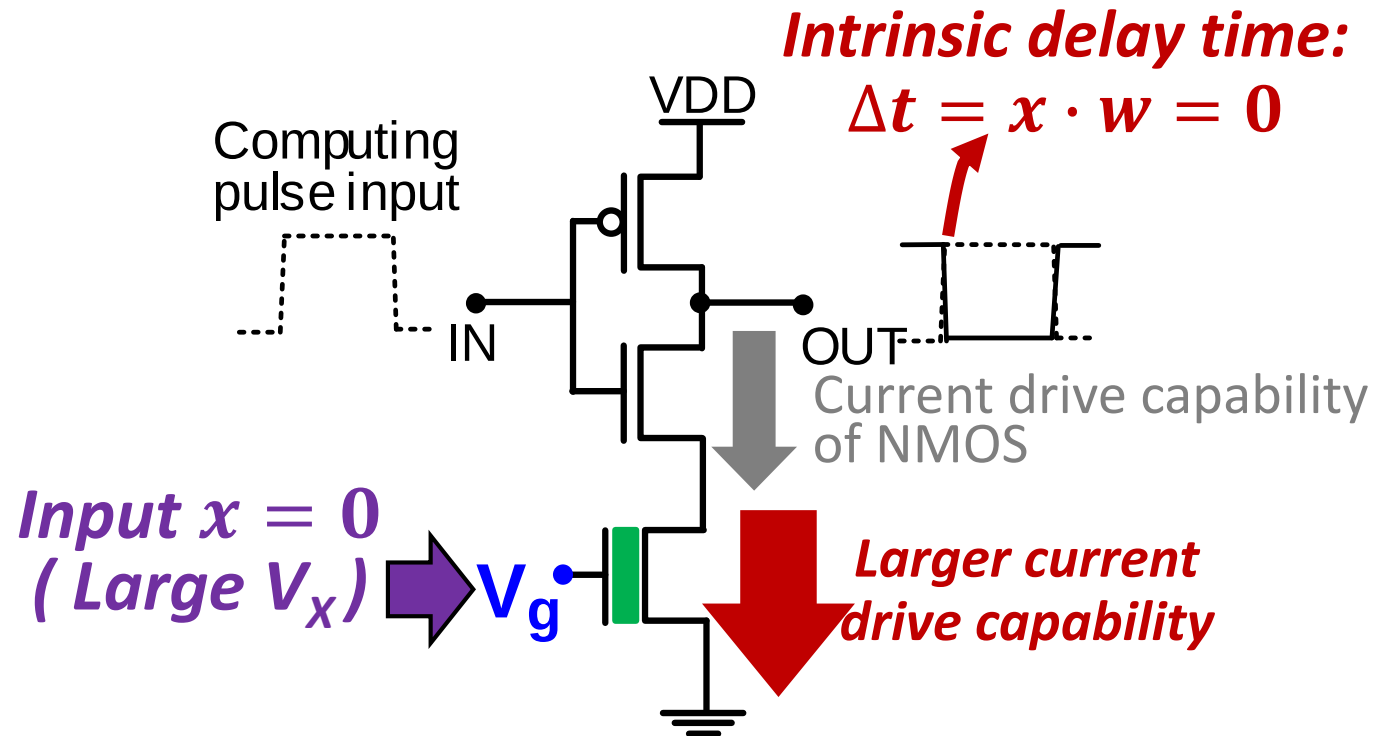
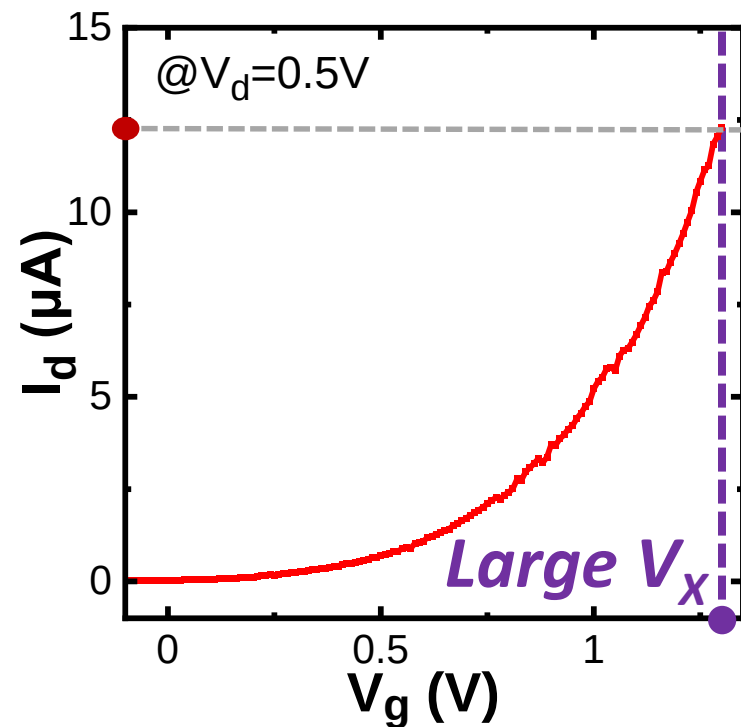
FeFET-based delay unit design (4/5)

- Input data $x = 1$: small V_x at non-destructive mode
 - The stored weight (w) determines pull-down current and delay time of falling edge



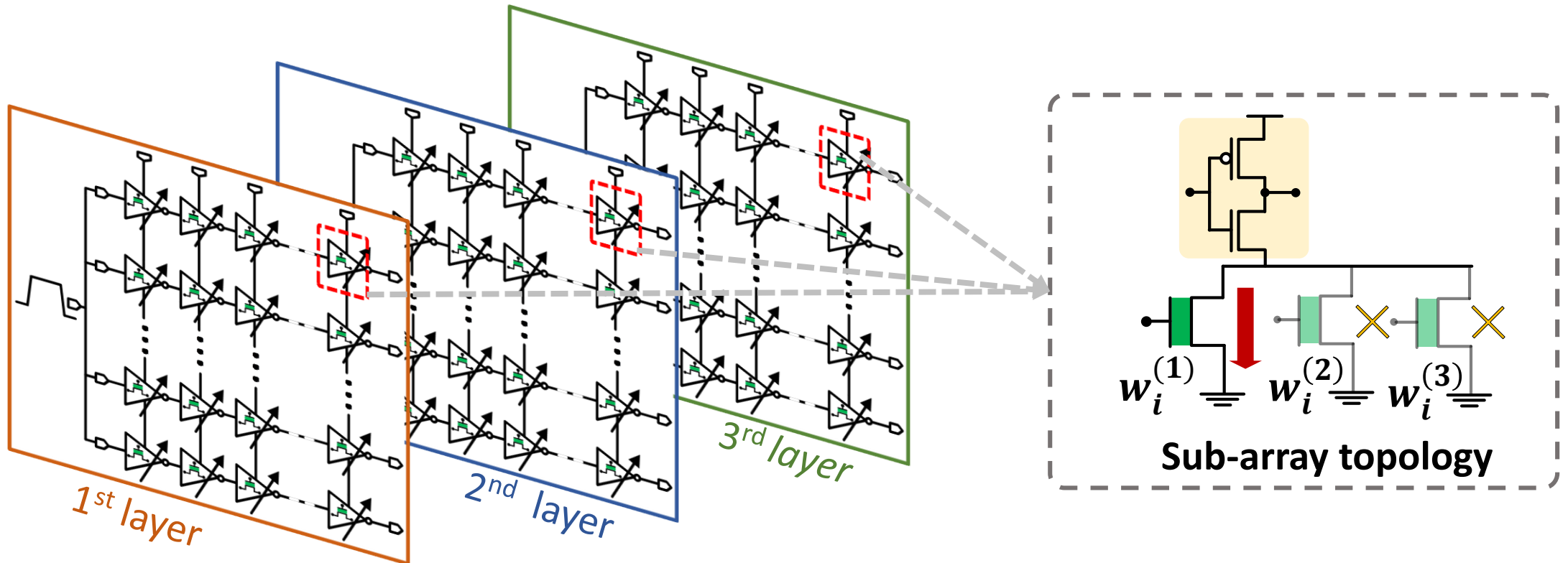
FeFET-based delay unit design (5/5)

- Input data $x = 0$: large V_x at non-destructive mode
 - FeFET is turned on with larger current drive capability than the pull-down NMOS
 - Small and weight-irrelevant intrinsic delay time



FeFET-based time-domain MAC array design

- Intrinsic MAC for $\vec{x} \cdot [W]$ with array of cascaded FeFET-based DUs (FE-DUs)
- The inverter of FE-DU can be reused by $[W]$ of different layers in neural network
→ **Further improve area-efficiency**

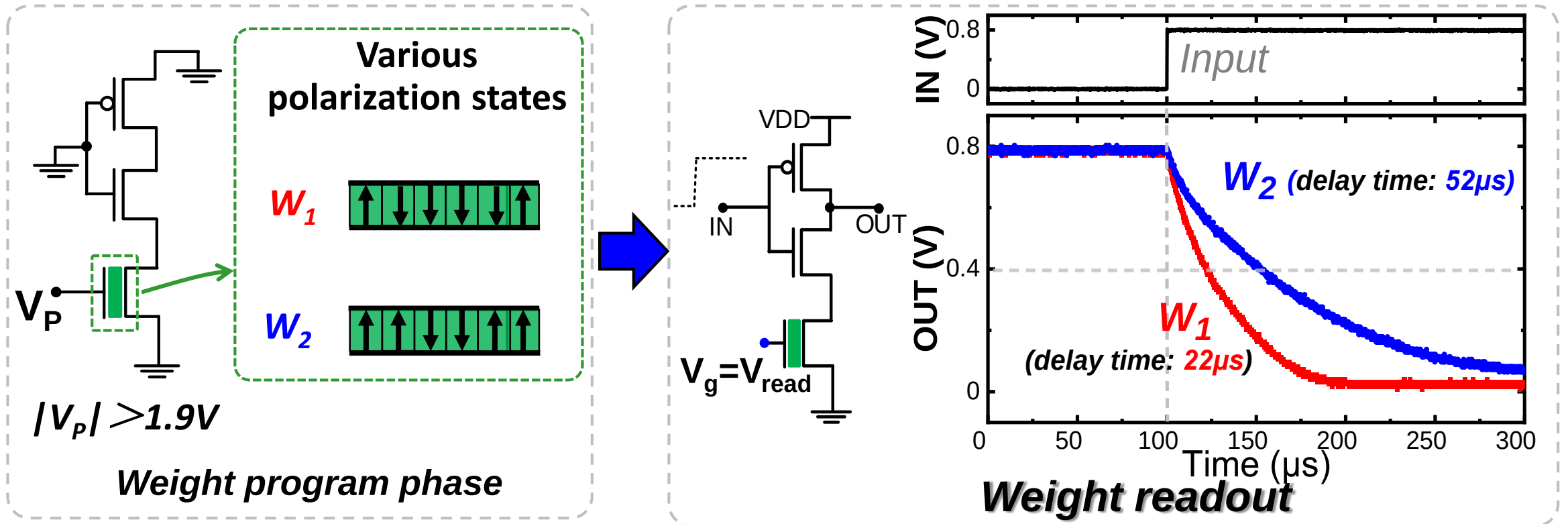


Outline

- Design of FeFET-based time-domain (TD) MAC
- **FeFET-based delay unit (DU) for TD MAC**
 - FeFET-based DU for local multiplication of MAC
 - Cascaded FeFET-based DUs for multiply-accumulate
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge application
- Summary

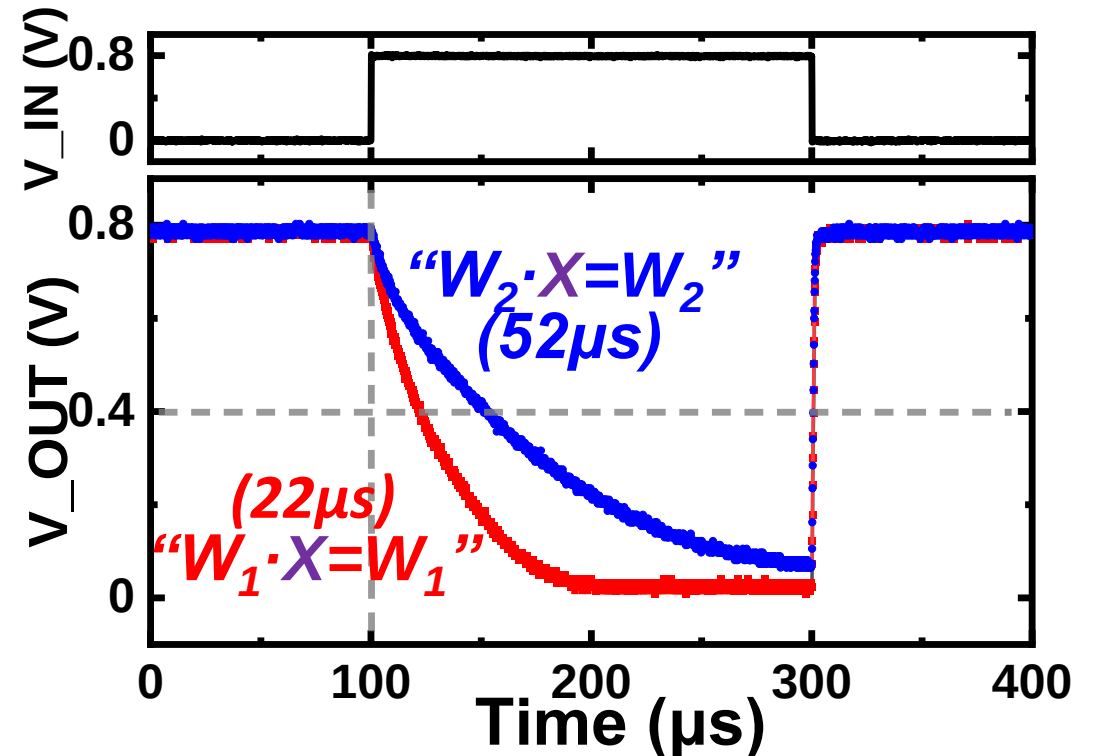
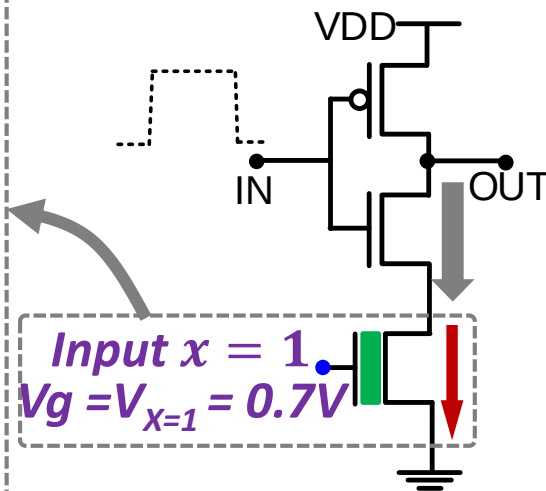
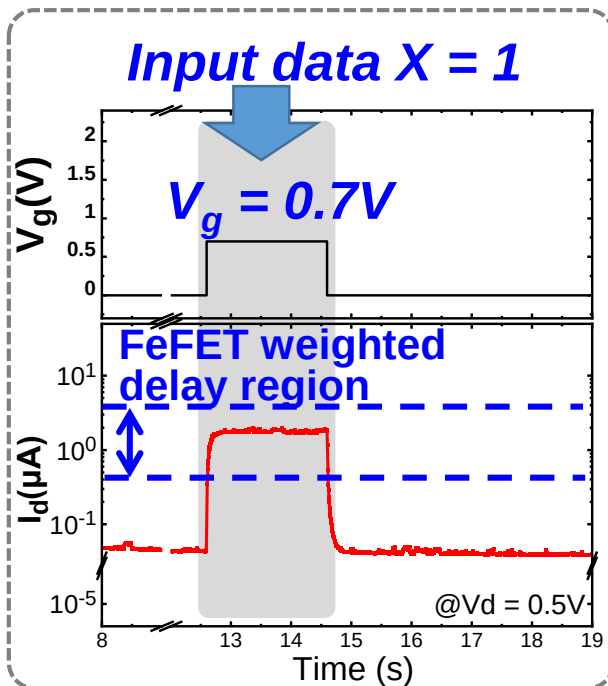
Experimental local TD multiplication (1/3)

- Program weight (w) with different program voltages (V_P)
 - Delay time of output falling edge is determined by different weight storages



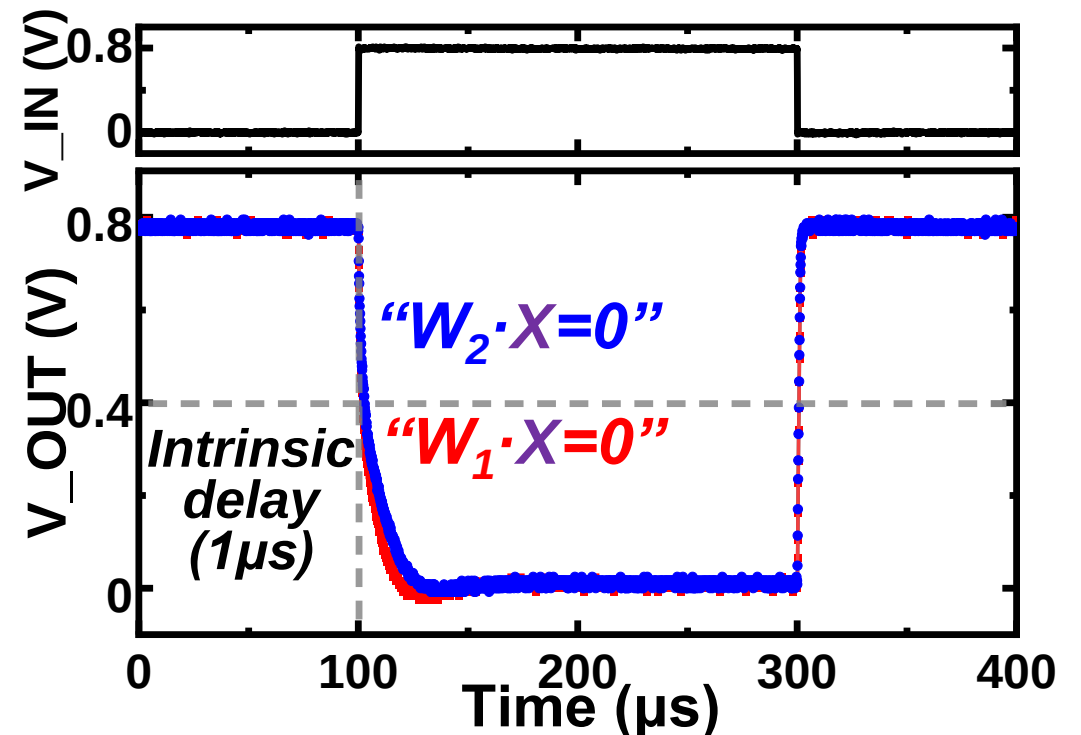
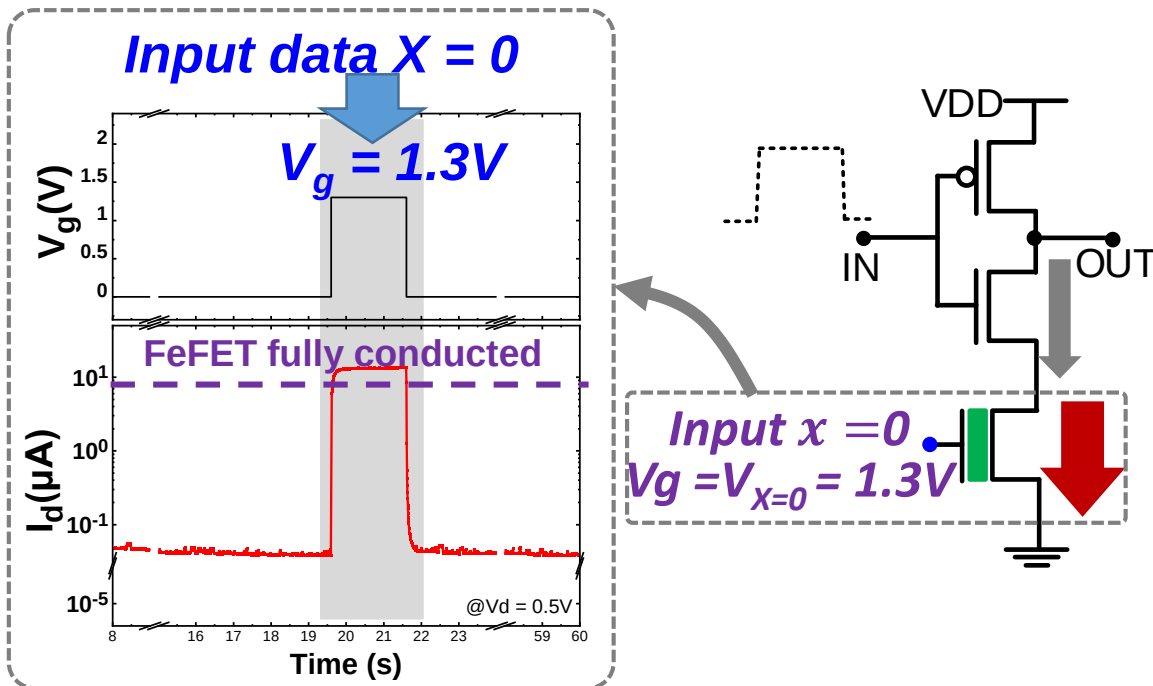
Experimental local TD multiplication (2/3)

- Multiplication phase with input data x
 - **Small V_x for $x = 1$** : pull-down current is modulated by weight stored in FeFET
 - Weighted propagation delays of falling edge



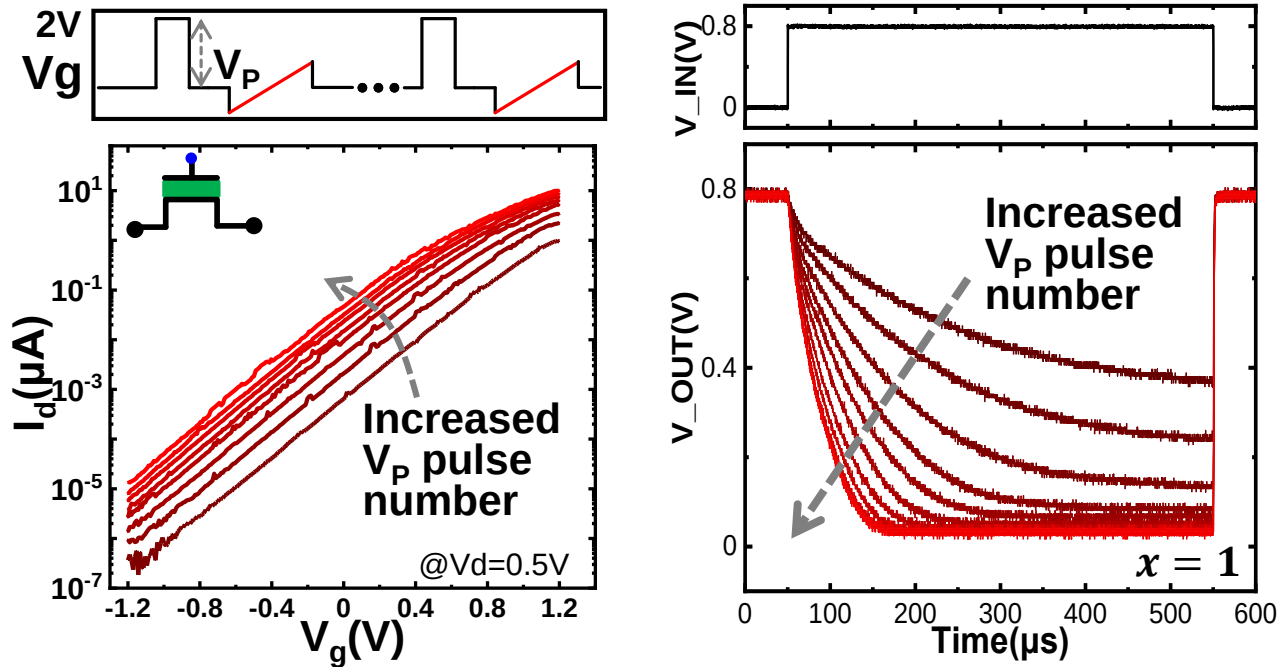
Experimental local TD multiplication (3/3)

- Multiplication phase with input data x
 - **Large V_x for $x = 0$:** sufficiently strong pull-down path at non-destructive mode
 - The time delays are unregulated by weights and determined by the small intrinsic delay of the inverter

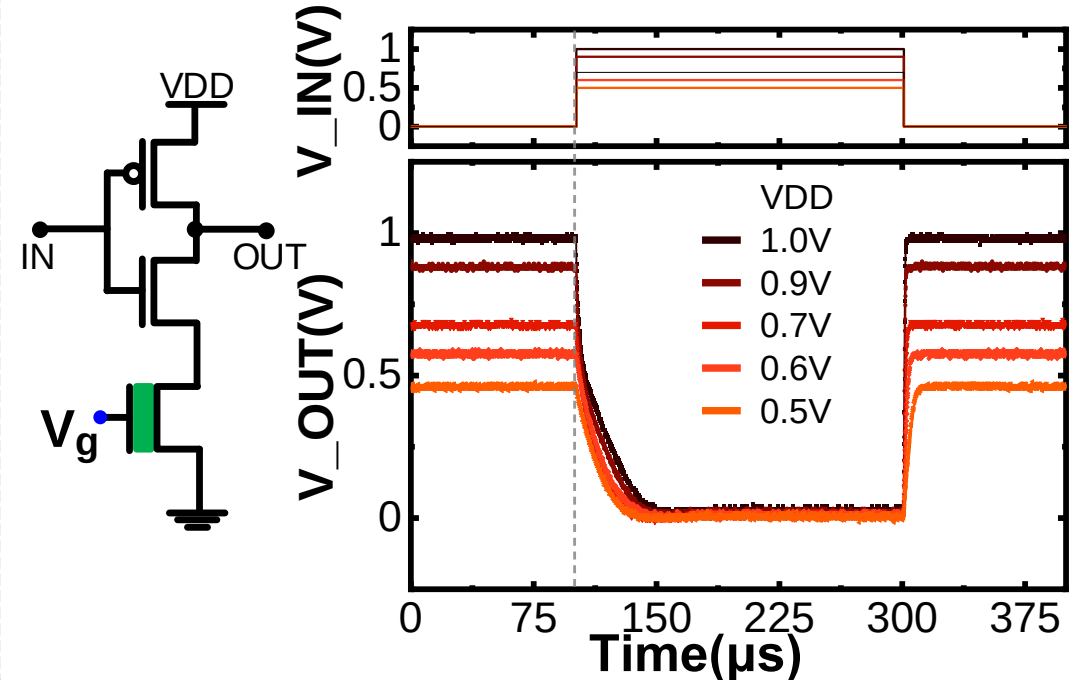


Multilevel weight storage & VDD scalability

- Multilevel weight storage of gradually modulated delay time



- The propagation process of FE-DU is consistent with VDD scaling



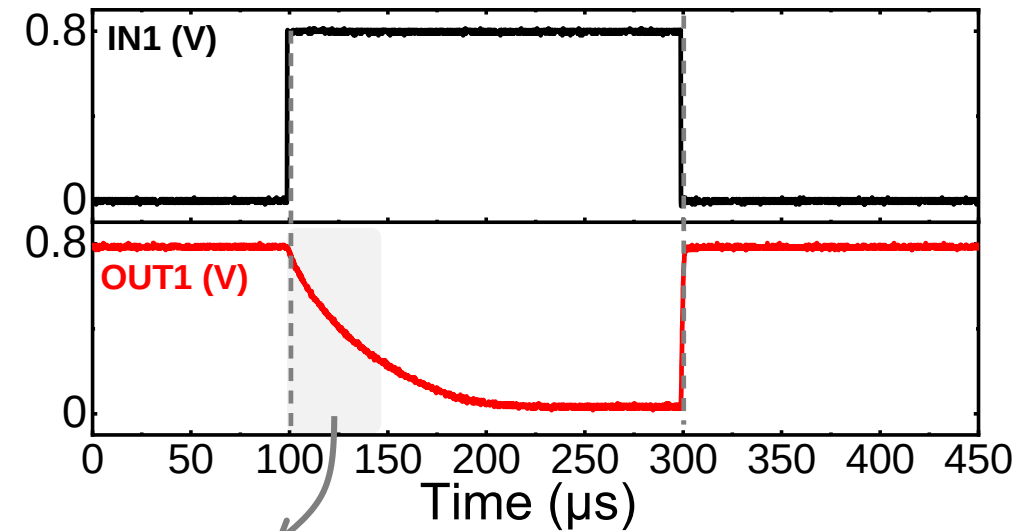
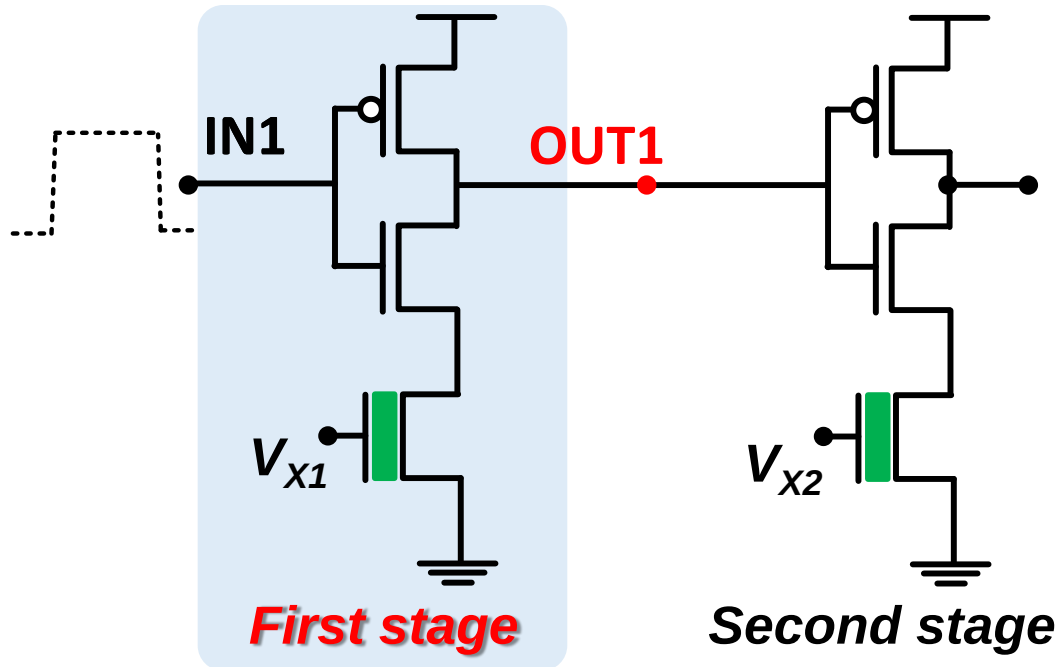
Outline

- Design of FeFET-based time-domain (TD) MAC
- **FeFET-based delay unit (DU) for TD MAC**
 - FeFET-based DU for local multiplication of MAC
 - Cascaded FeFET-based DUs for multiply-accumulate
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge application
- Summary

Dual-edge operation of cascaded FE-DUs (1/3)

- First stage

- The polarity of OUT1 is contrary to input pulse due to the inverter
- Falling edge of OUT1: with modulated delay due to local multiplication ($x_1 \cdot w_1$)
- Rising edge of OUT1: without delay due to pull-up path of PMOS

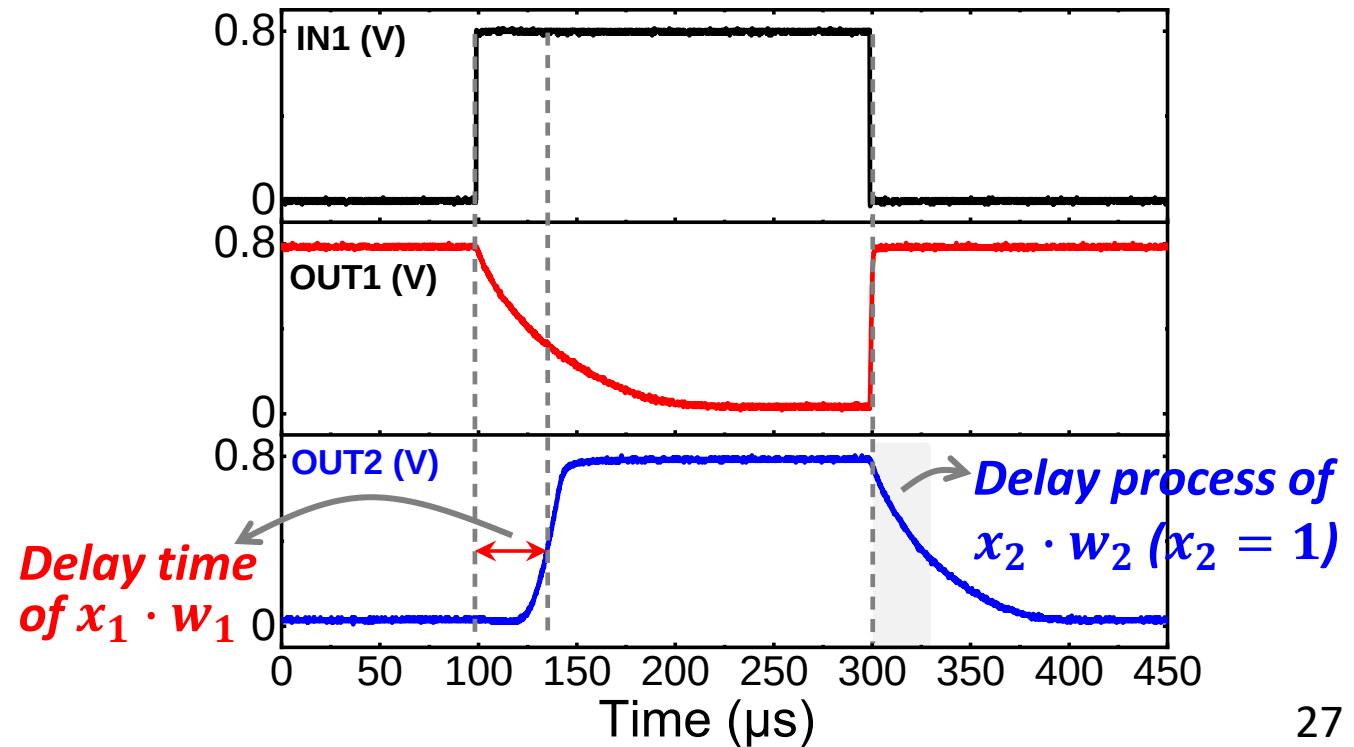
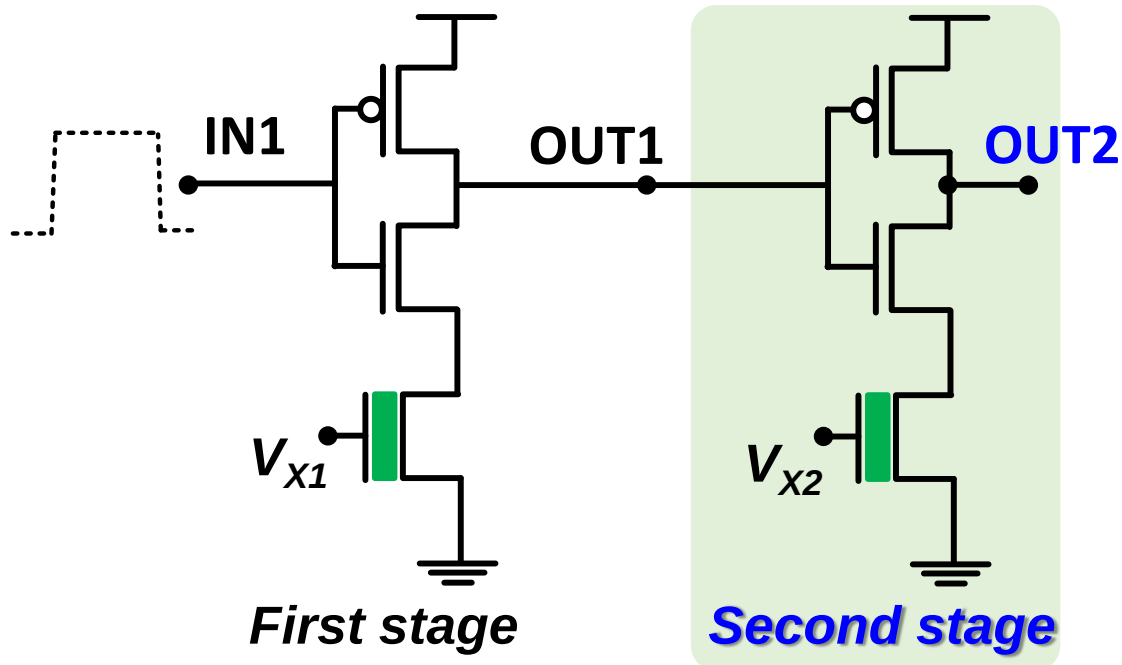


*Delay process of
 $x_1 \cdot w_1$ ($x_1 = 1$)*

Dual-edge operation of cascaded FE-DUs (2/3)

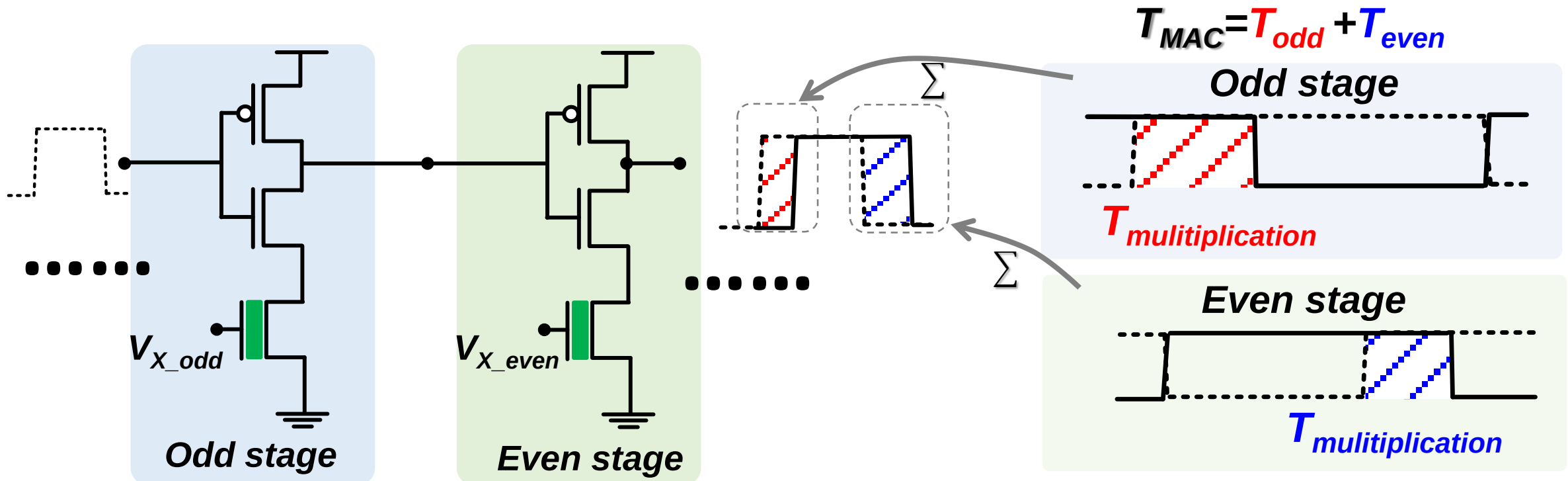
- Second stage

- The polarity of OUT2 is contrary to OUT1, and the same to input pulse
- Rising edge of OUT2: with delay time ($\Delta t = x_1 \cdot w_1$) compared with initial input
- Falling edge of OUT2: with modulated delay due to local multiplication ($x_2 \cdot w_2$)



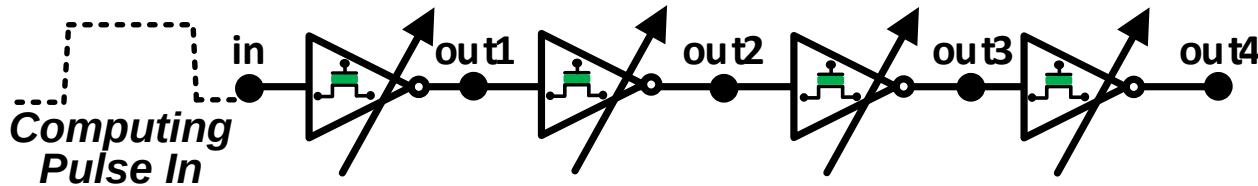
Dual-edge operation of cascaded FE-DUs (3/3)

- Both dual-edge transitions of computing pulse contribute to TD MAC
- Time-domain MAC result (T_{MAC})
 - Sum of delay time of front edge in odd stages and back edge in even stages



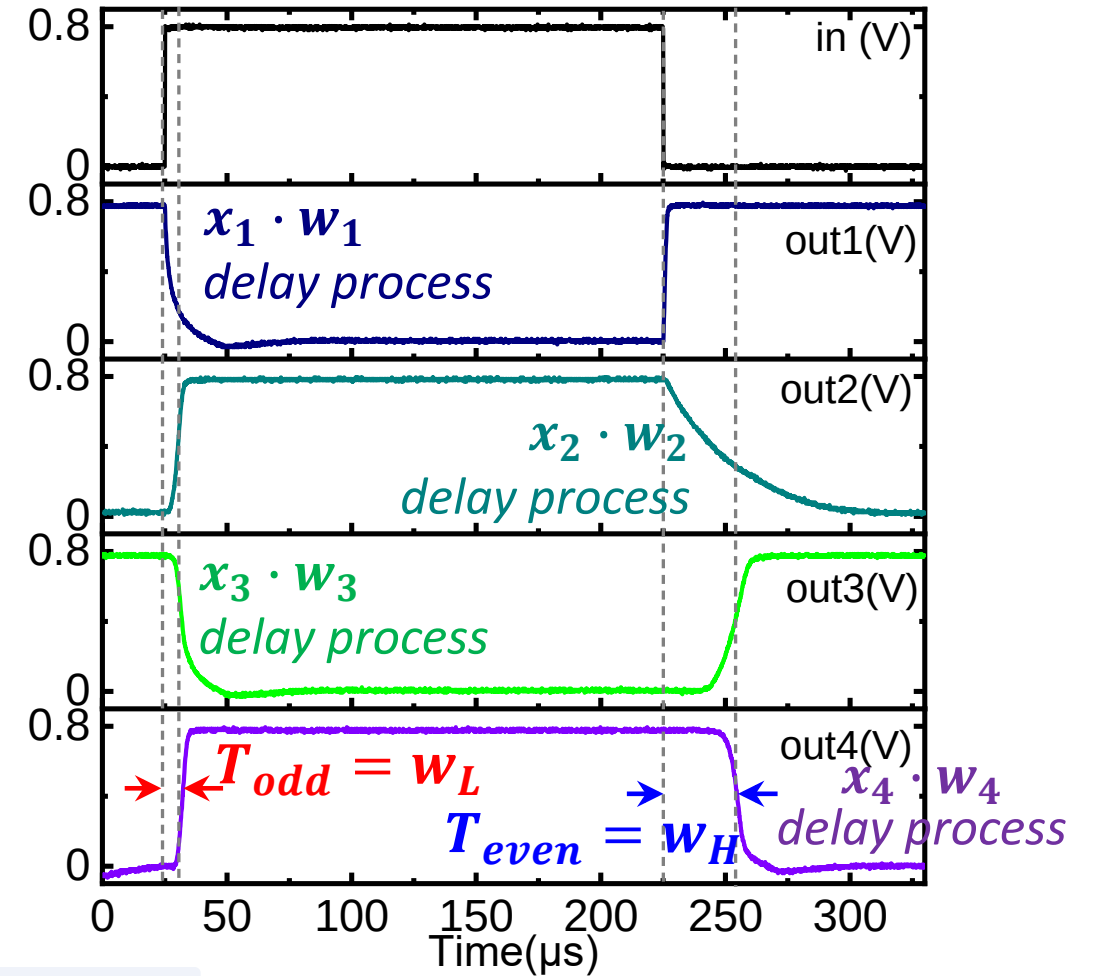
Cascade of multiple FeFET-based DUs

- Experimental results of cascaded four stages of FeFET-based DUs



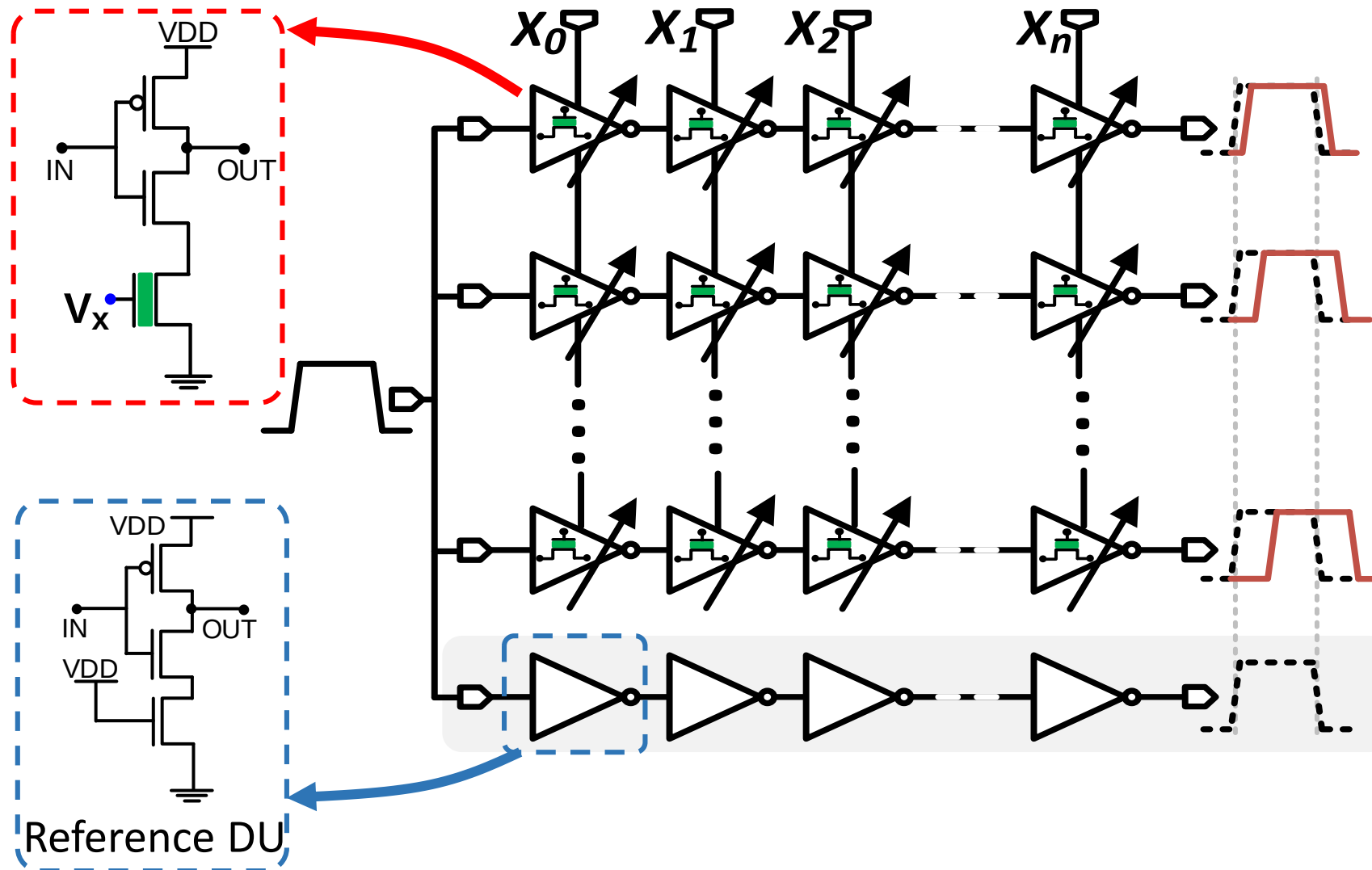
	Stage #1	Stage #2	Stage #3	Stage #4
Weight w	$W_1=W_L$	$W_2=W_H$	$W_3=W_L$	$W_4=W_H$
Input x	1	1	0	0
$x \cdot w$	W_L	W_H	0	0

W_L for small delay; W_H for large delay



$$T_{MAC} = T_{odd} + T_{even} = x_1 w_1 + x_3 w_3 + x_2 w_2 + x_4 w_4$$

FeFET-based DU array for highly-parallel MAC

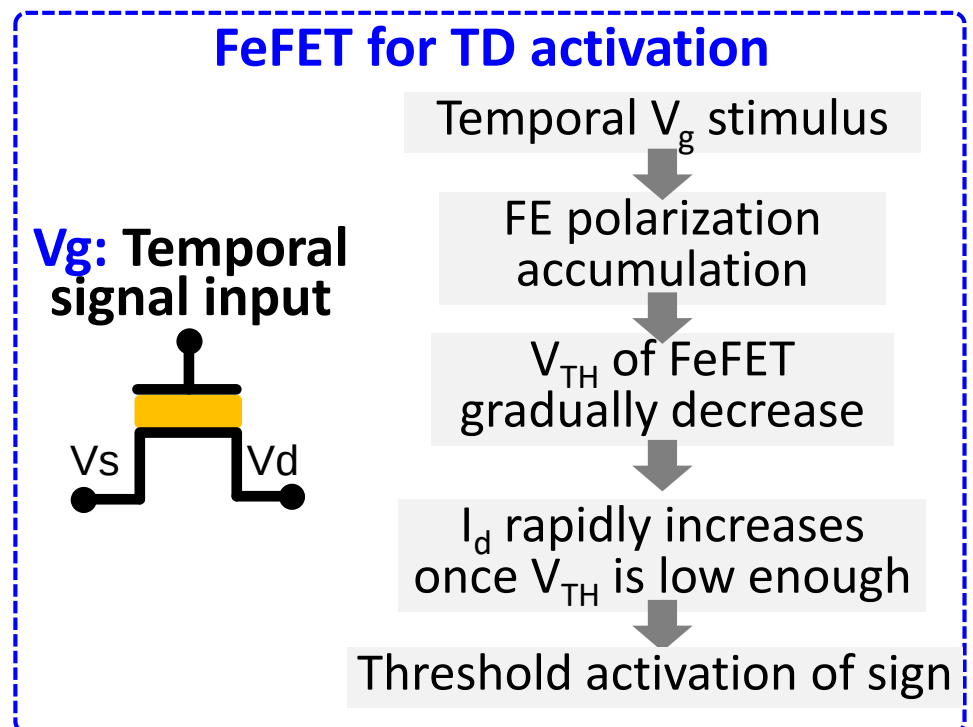
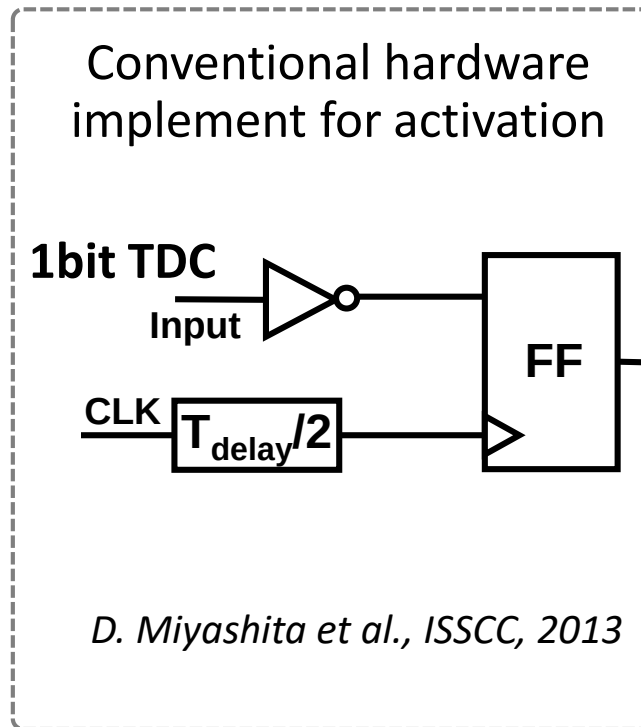
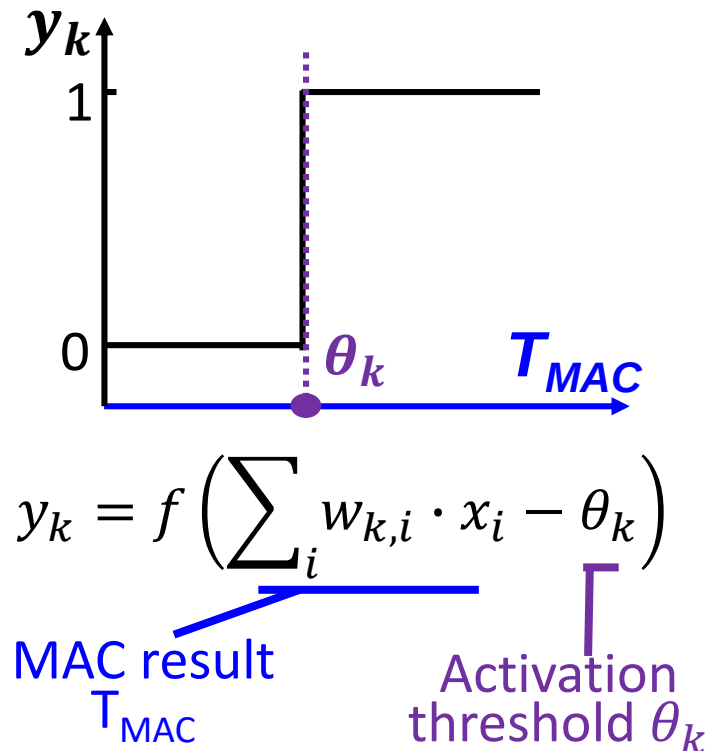


Outline

- Design of FeFET-based time-domain (TD) MAC
- FeFET-based delay unit (DU) for TD MAC
- **FeFET-based TD activation**
- FeFET-based TD nvCIM macro for edge application
- Summary

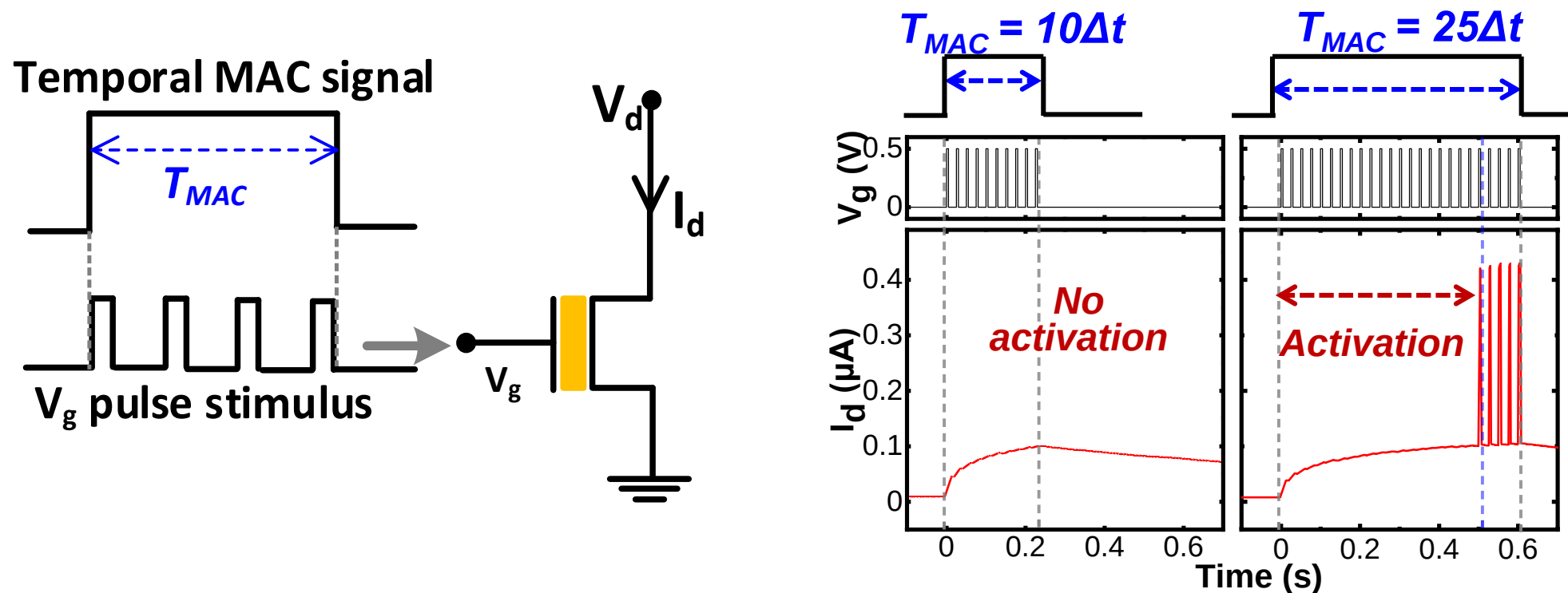
Design of FeFET-based TD activation

- Nonlinear sign activation function for time-domain MAC results (T_{MAC})
 - Simplified threshold activation with adjustable temporal threshold θ_k
- Time-domain activation with only **1FeFET**



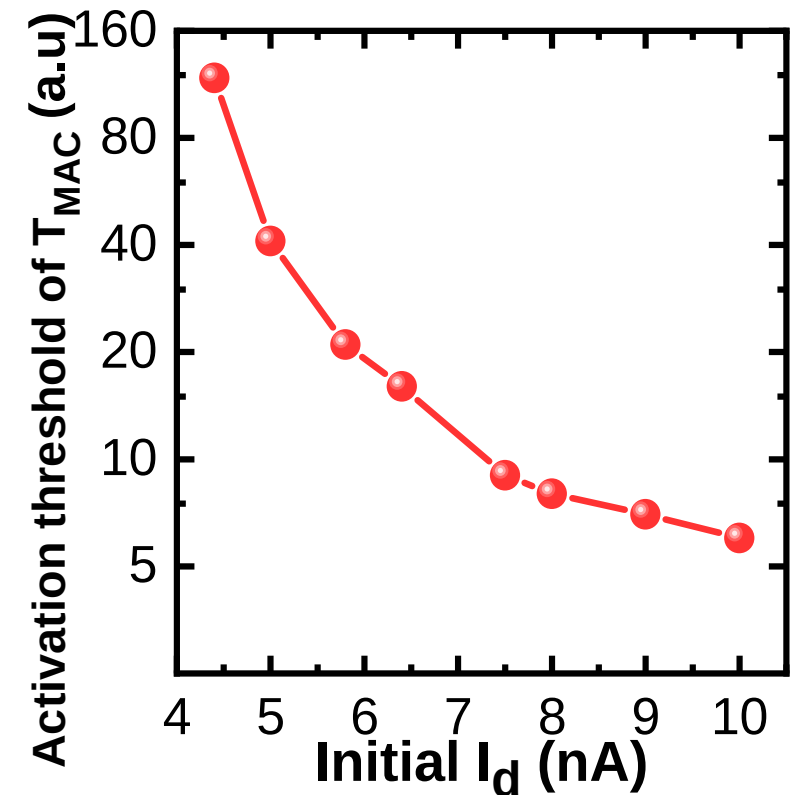
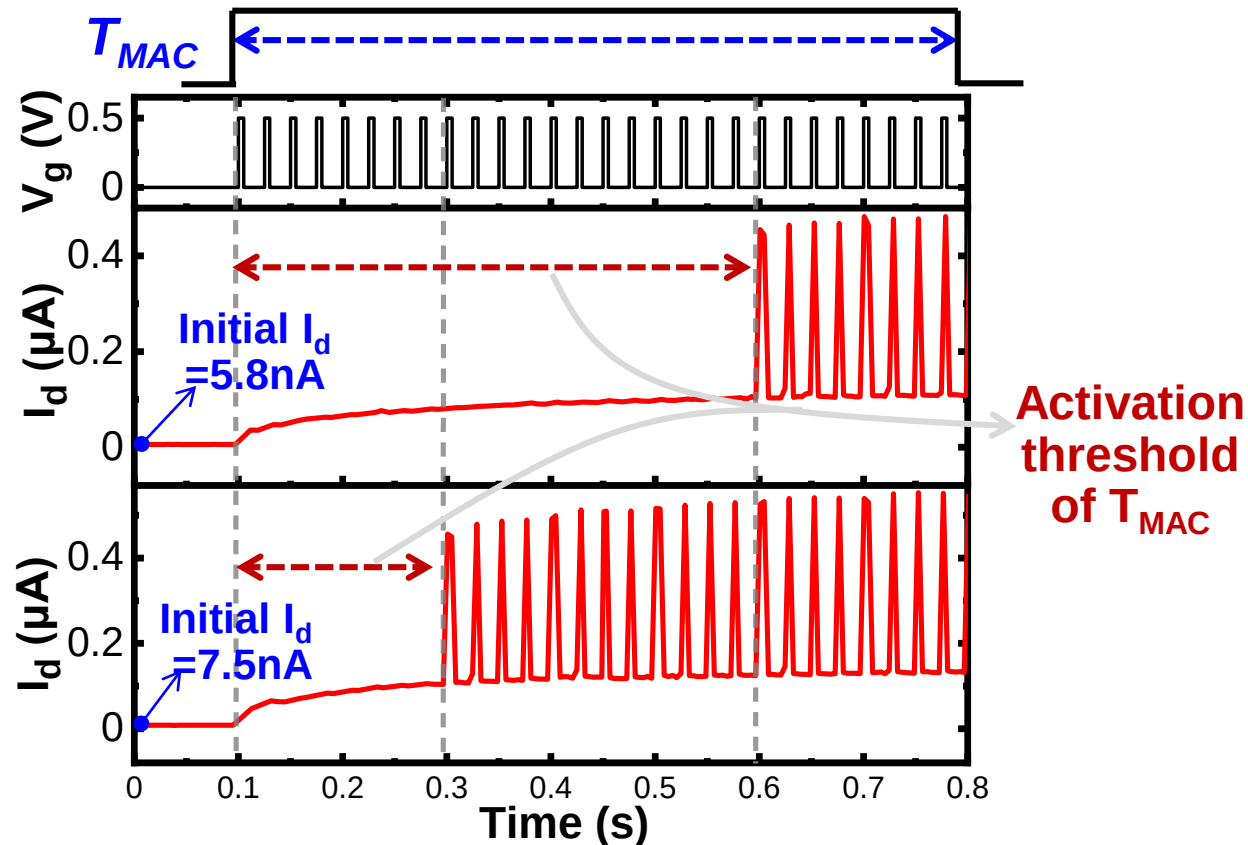
FeFET-based nonlinear TD sign activation (1/3)

- Temporal threshold activation behavior
 - V_g pulse stimulus is applied to the gate of FeFET during T_{MAC}
 - Large enough T_{MAC} triggers output activation due to polarization accumulation



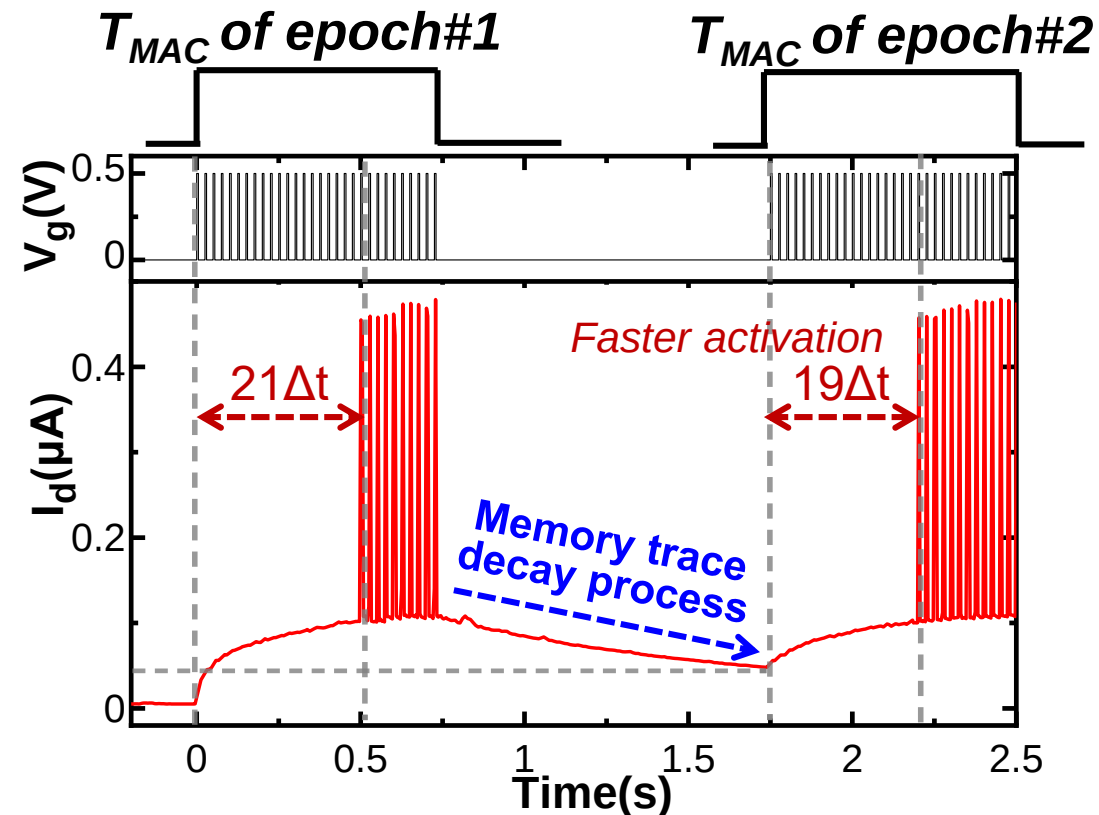
FeFET-based nonlinear TD sign activation (2/3)

- Threshold activation with adjustable temporal threshold
 - The time required for triggering output activation can be adjusted by modulating the initial base state of the FeFET



FeFET-based nonlinear TD sign activation (3/3)

- Threshold activation with memory trace behavior
 - Short-term relaxation for the faster activation with immediate input due to depolarization field induced time-dependent polarization degradation

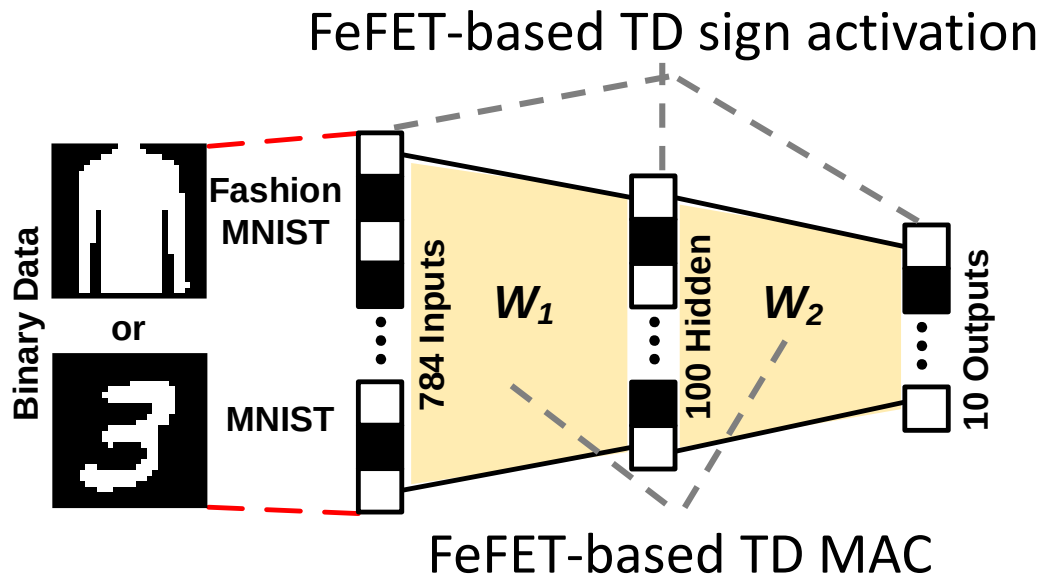


Outline

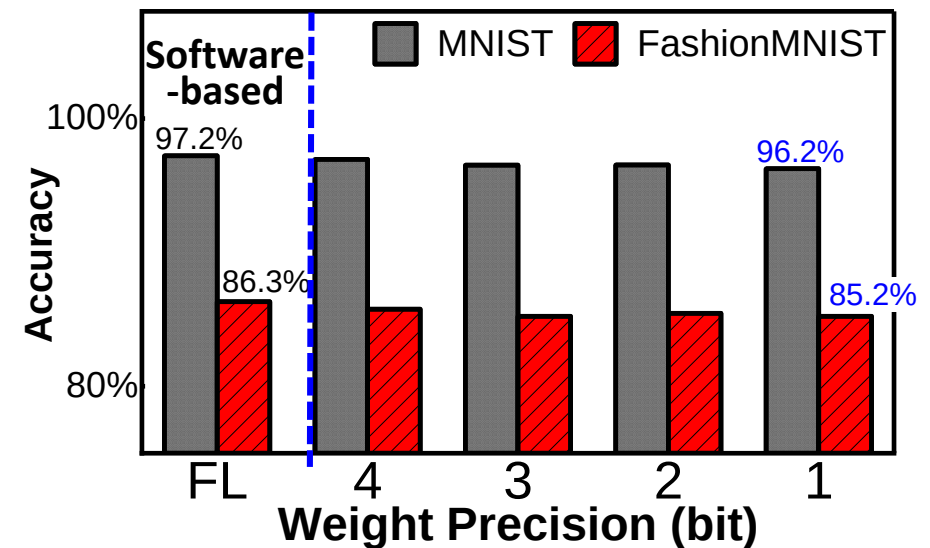
- Design of FeFET-based time-domain (TD) MAC
- FeFET-based delay unit (DU) for TD MAC
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge applications
 - Pattern recognition & reinforcement learning
 - MAC array performance evaluation
- Summary

Pattern recognition with FeFET-based nvCIM

- ANN structure based on proposed FeFET-based nvCIM macro
 - TD MAC based on FE-DU with multilevel weight & sign activation based on FeFET
- **High accuracy** inference for typical pattern recognition



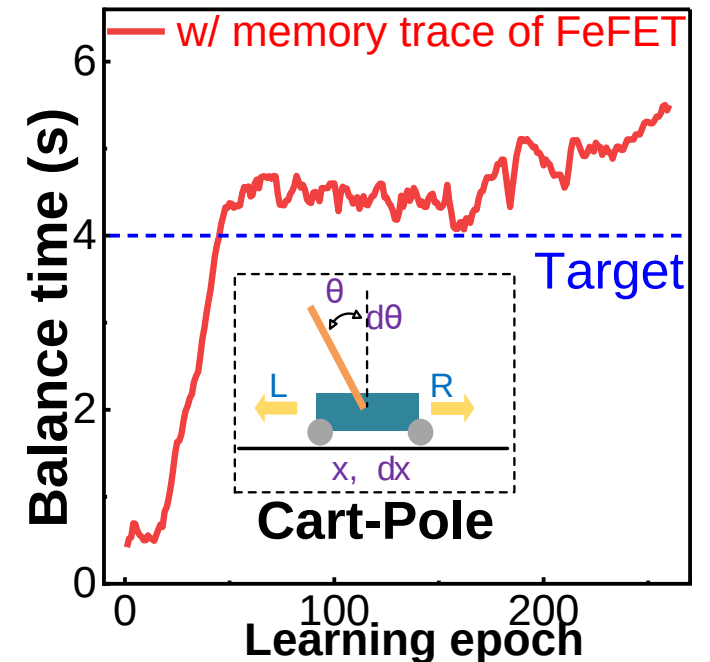
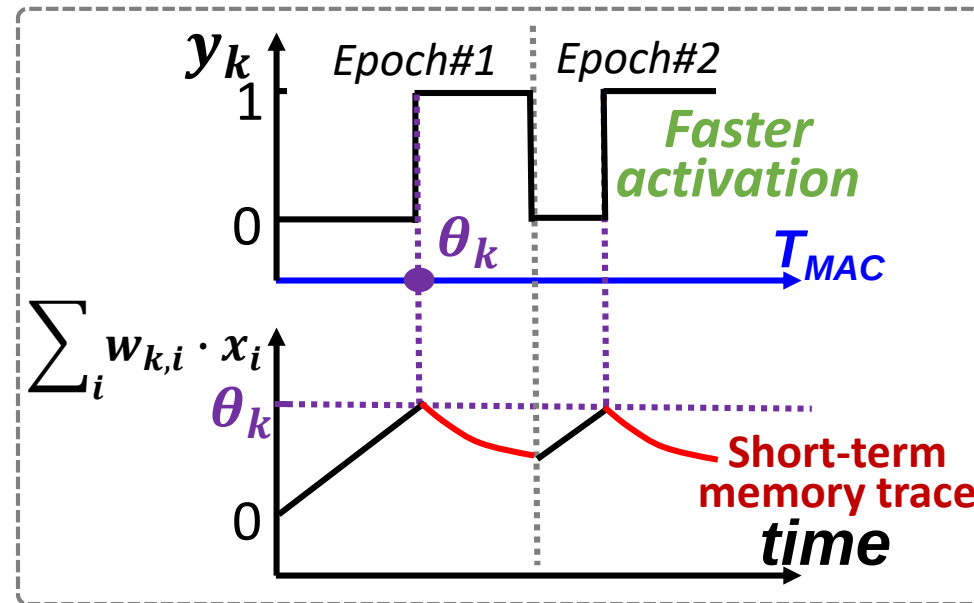
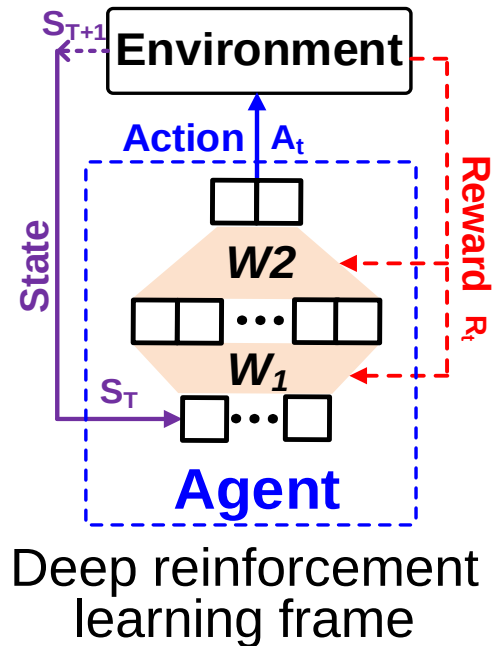
Artificial neural network (ANN) structure



96.2% @MNIST,
85.2% @Fashion-MNIST

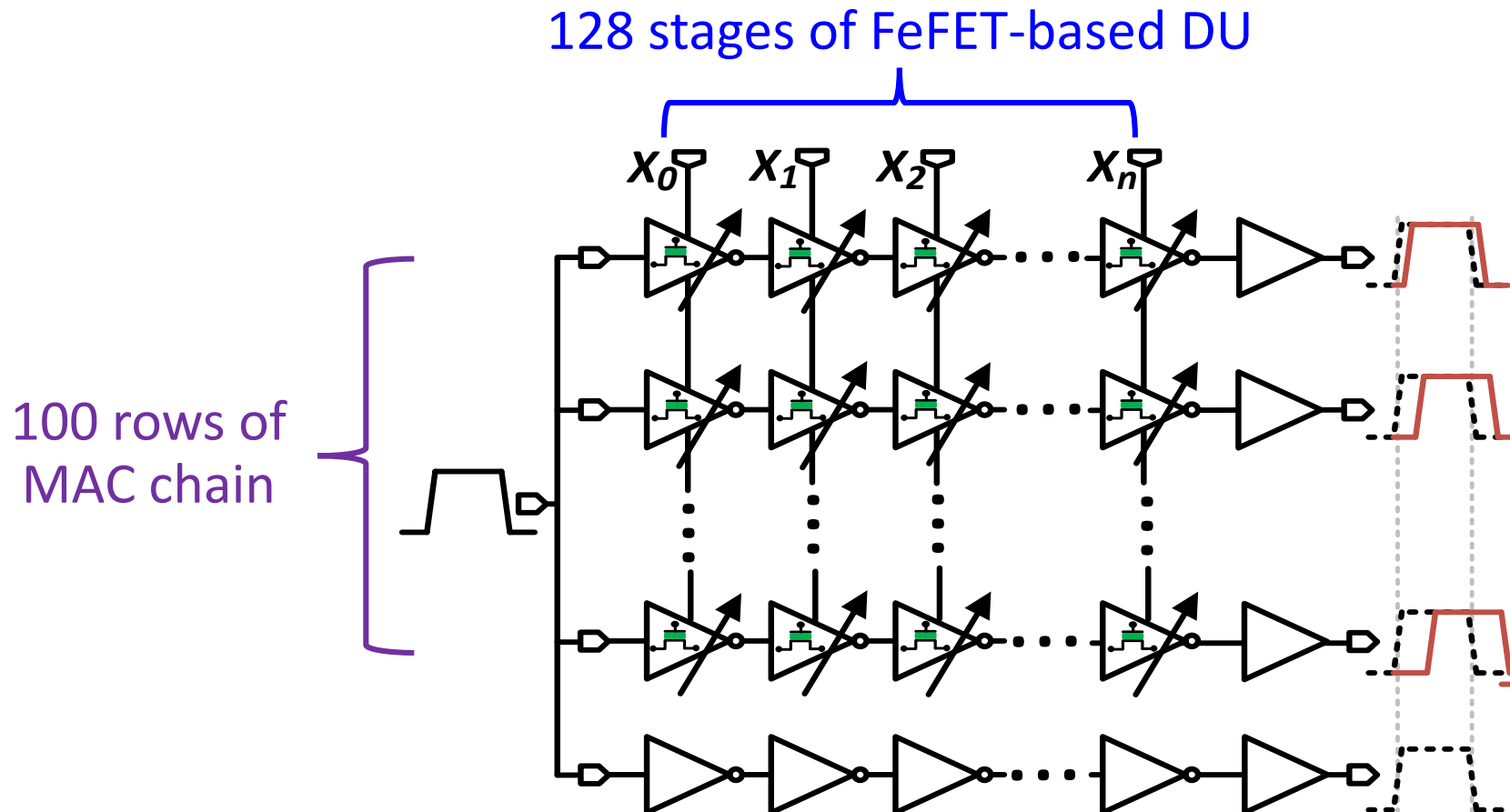
Reinforcement learning with FeFET-based nvCIM

- The agent of reinforcement learning is based on FeFET-based ANN
 - Benefit from intrinsic short-term memory trace of FeFET-based activation
 - More energy- and area-efficiency than conventional capacitor-based method
- Accelerated convergence rate of learning process



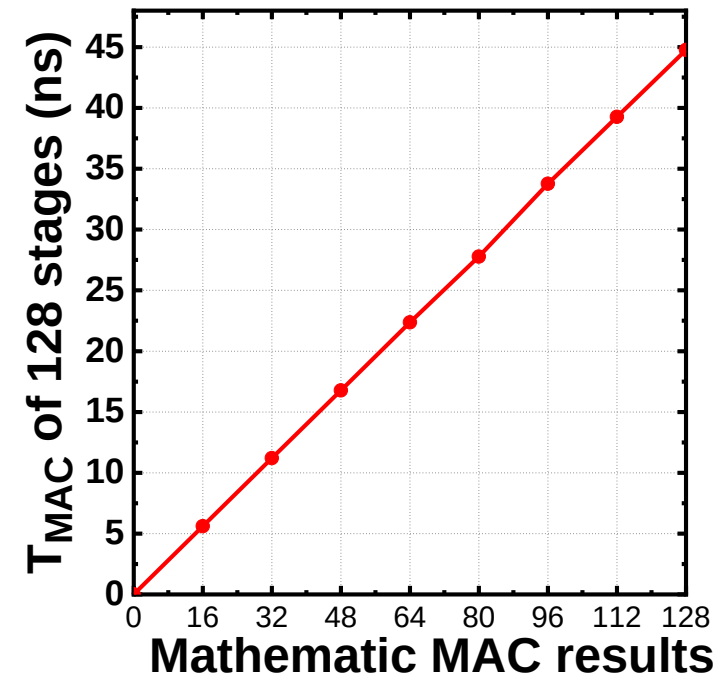
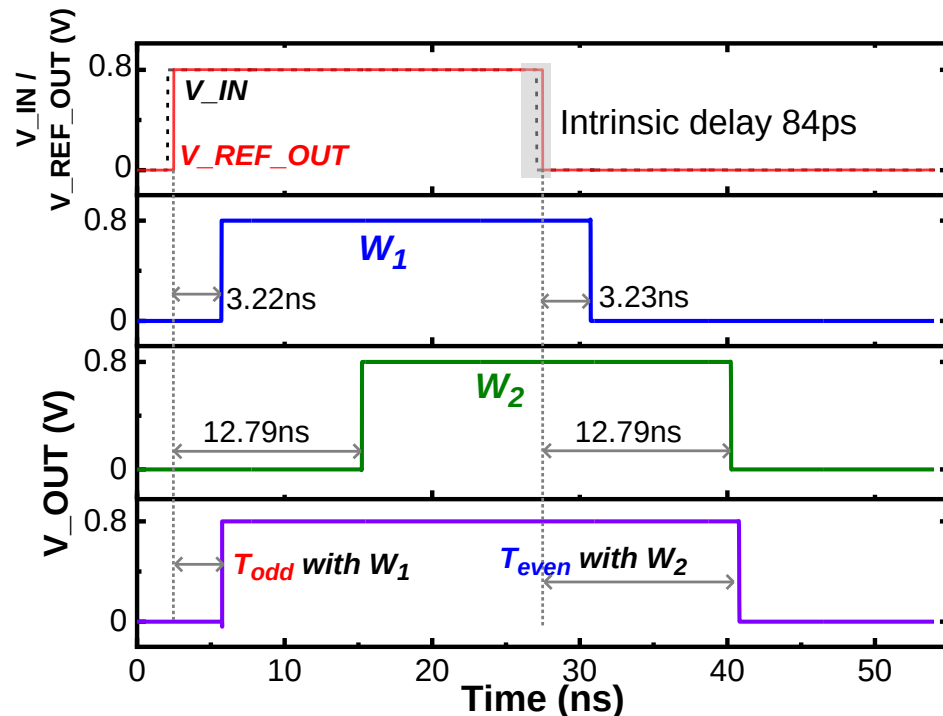
MAC array performance evaluation

- FeFET-based MAC array based on developed Fe-FinFET model



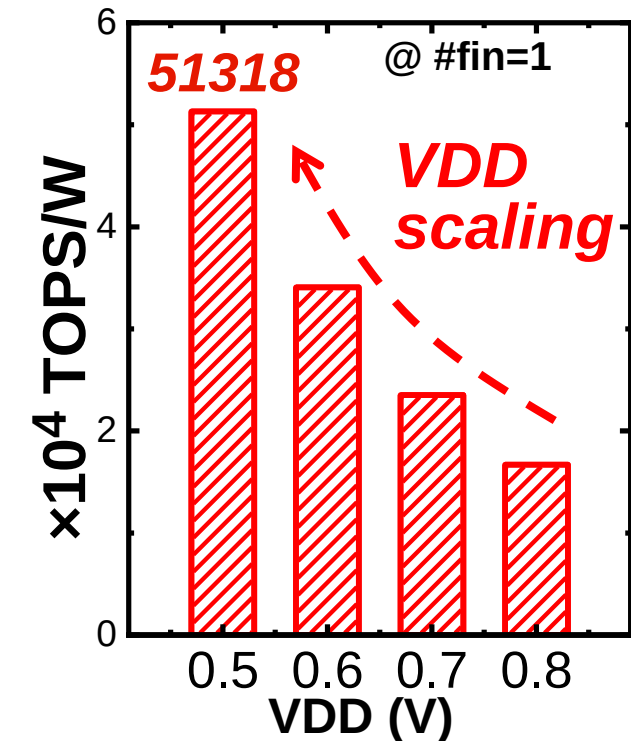
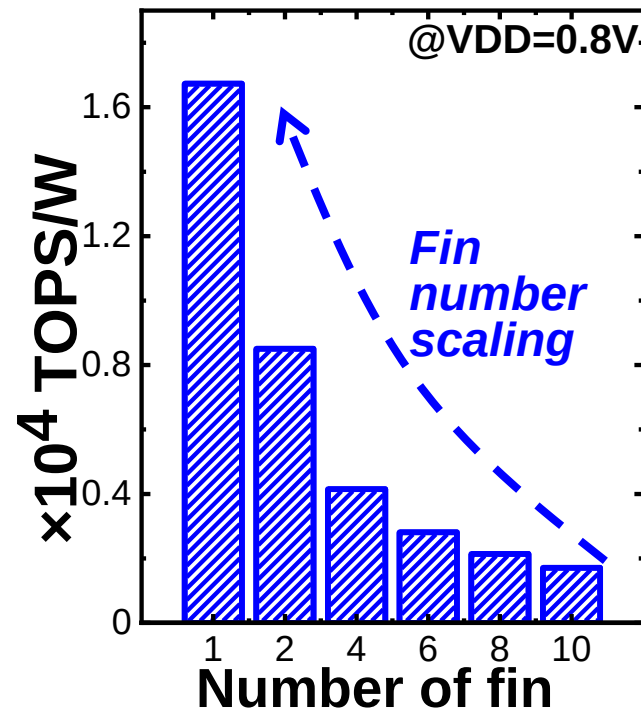
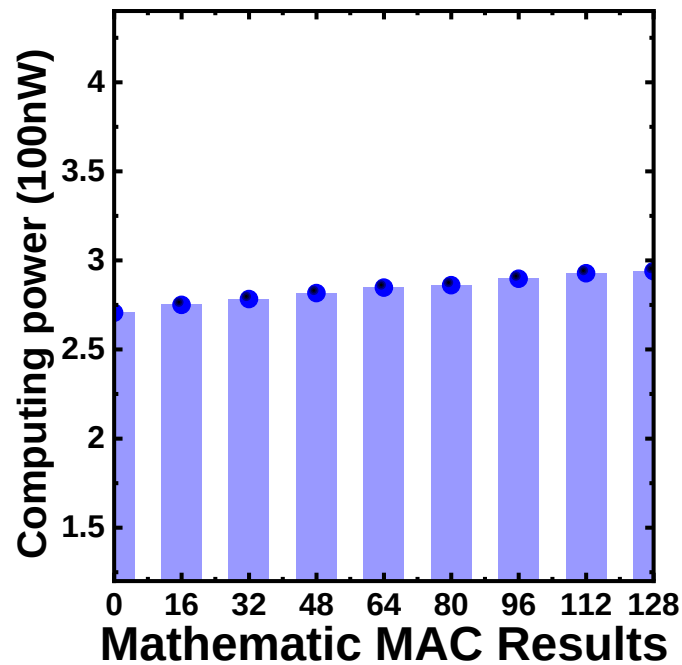
MAC array performance evaluation

- FeFET-based time-domain MAC computing
 - Intrinsic asynchronous serial MAC with dual-edge operation of various weights can be implemented with large signal dynamic range in time domain
 - Time-domain computing results show good linearity with mathematic MAC results



MAC array performance evaluation

- Energy efficiency of proposed FeFET-based TD MAC
 - MAC results independent computing power due to consistent toggle activity
 - Improved energy efficiency with fin number decreasing and VDD scaling



Benchmark of the proposed FeFET-based time-domain nvCIM with other reported works

	Ref. [1]	Ref. [2]	Ref. [3]	Ref. [4]	This work
Analog Domain	Voltage	Time			Time
Weight memory cell	FeFET	8T-SRAM	SRAM	6T-SRAM	Fe-FinFET
NVM	Yes	No			Yes
Cell size	1T+1R	> 19T	> 55T	12T	3T (1FeFinFET+2T)
Multi-bit cell	--	No			Yes
Activation	ADC	Digital logic blocks			One FeFET
Supply voltage	0.5-1.2V	0.6V	1.0V	0.9V	0.5V
ANN application	--	Pattern recognition			Pattern recognition & Reinforcement learning
MAC Energy efficiency (TOPS/W)	1.3×10^4	119.7	48.5	716	5×10^4 TOPS/W

[1] T. Soliman et al., *IEDM*, pp. 29.2.1, 2020.

[2] J. Yang et al., *ISSCC*, pp. 394, 2019.

[3] D. Miyashita et al., *JSSC*, pp. 2679, 2017.

[4] J. Song et al., *TCAS-I*, pp. 3377, 2021.

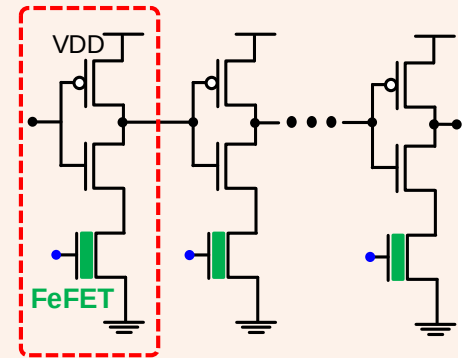
Outline

- Design of FeFET-based time-domain (TD) MAC
- FeFET-based delay unit (DU) for TD MAC
- FeFET-based TD activation
- FeFET-based TD nvCIM macro for edge application
- **Summary**

Summary

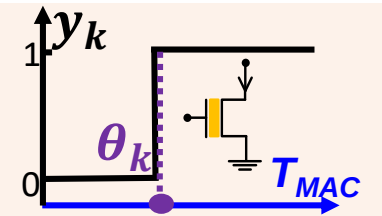
Fe-FinFET-based delay unit with $2T+1$ FeFET for time-domain nvCIM

- FeFET: *non-volatile polarization* for weight storage, *controllable switch* for input modulation **simultaneously**, realizing **local multiplication**
- **Multilevel weight** storage and **VDD scaling**
- Cascaded FE-DUs for time-domain MAC with **dual-edge operation** for further improving energy efficiency



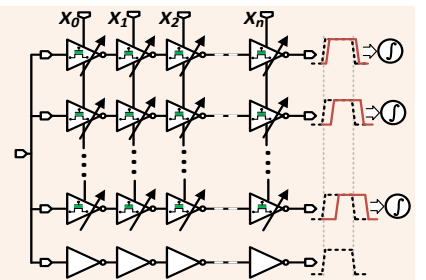
Time-domain activation with only 1 FeEFT

- *Time-dependent FE switching* for TD activation with **memory trace**



Edge AI based on proposed FeFET-based time-domain nvCIM

- High-accuracy pattern recognition & accelerated reinforcement learning
- **High energy efficiency and ultra-low hardware cost**



Thank you for your attention!